

Supplement Section B: Inferential statistics

In this document;

- **B1:** Comprehensive models with exploratory variables and variable exclusion process
- **B2:** Unsplit models (without stimulus type separation)
- **B3:** Complete models including control condition
- **B4:** Factorial 2x2 models
- **B5:** Normality assessments and Model assumptions
- **B6:** Power Analysis
- **B7:** Expected vs. Observed ratings of item photographs

In the .Pdf output file you may find only the outputs generated by the analysis code, in order to keep the supplementary files as concise as possible. However the full code is available on <https://github.com/Rezvani-S/Costly-Beauty-ShoeStudy.git> (<https://github.com/Rezvani-S/Costly-Beauty-ShoeStudy.git>)

1. Required Packages

2. Data Import and Preparation

PART I: Supplement B1: Comprehensive models with exploratory variables and variable exclusion process

3. Data Cleaning and Preparation

4. Comprehensive Attractiveness Model

A comprehensive linear mixed-effects model for attractiveness was constructed, including all potential variables:

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Rating ~ Condition + sex + monthlyincome + merged_score_zscore +
## Stimuli_Type + (1 | block/Model) + (1 | code)
## Data: body_df_attract
##
## REML criterion at convergence: 47192.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.9177 -0.6141 -0.0090  0.6115  4.3914
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## code        (Intercept) 1.718e+00 1.3108014
## Model:block (Intercept) 6.051e-01 0.7778898
## block       (Intercept) 3.105e-08 0.0001762
## Residual                    1.784e+00 1.3355910
## Number of obs: 13435, groups: code, 275; Model:block, 54; block, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      5.679e+00  2.242e-01 3.054e+02  25.325 < 2e-16 ***
## ConditionEconomy Flat -1.474e-01  3.643e-02 1.311e+04  -4.045 5.26e-05 ***
## ConditionEconomy Heel  2.179e-02  3.643e-02 1.311e+04   0.598 0.549834
## ConditionLuxury Flat -1.257e-01  3.643e-02 1.311e+04  -3.450 0.000562 ***
## ConditionLuxury Heel -1.552e-01  3.645e-02 1.311e+04  -4.259 2.06e-05 ***
## sexMale           -1.580e-01  1.842e-01 2.659e+02  -0.858 0.391641
## monthlyincome.L     4.441e-01  4.196e-01 2.706e+02   1.059 0.290753
## monthlyincome.Q      1.810e-01  3.587e-01 2.692e+02   0.505 0.614213
## monthlyincome.C      4.240e-01  2.874e-01 2.709e+02   1.475 0.141284
## merged_score_zscore -5.405e-03  1.011e-01 2.698e+02  -0.053 0.957388
## Stimuli_TypeVideo  -5.135e-01  2.305e-02 1.311e+04 -22.281 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) CndtEF CndtEH CndtLF CndtLH sexMal mnth.L mnth.Q mnth.C
## CndtnEcnmyF -0.081
## CndtnEcnmyH -0.081  0.500
## CndtnLxryFl -0.081  0.500  0.500
## CndtnLxryHl -0.081  0.500  0.500  0.500
## sexMale     -0.362  0.000  0.000  0.000  0.000
## mnthlyncm.L  0.688  0.000  0.000  0.000  0.000 -0.040
## mnthlyncm.Q  0.339  0.000  0.000  0.000  0.000  0.087  0.747
## mnthlyncm.C  0.133  0.000  0.000  0.000  0.000  0.050  0.297  0.602
## mrgd_scr_zs -0.107  0.000  0.000  0.000  0.000  0.375  0.015  0.012 -0.055
## Stiml_TypVd -0.051  0.000  0.000  0.000  0.000  0.000  0.000  0.000  0.000
##              mrgd__
## CndtnEcnmyF
## CndtnEcnmyH
## CndtnLxryFl
## CndtnLxryHl
## sexMale
## mnthlyncm.L

```

```
## mnthlyncm.Q
## mnthlyncm.C
## mrgd_scr_zs
## Stiml_TypVd 0.000
```

ANOVA for Fixed Effects in Attractiveness Model

```
## Type III Analysis of Variance Table with Satterthwaite's method
##
##          Sum Sq Mean Sq NumDF   DenDF  F value    Pr(>F)
## Condition      78.01   19.50     4 13112.0  10.9337 7.53e-09 ***
## sex             1.31    1.31     1   265.9   0.7362  0.3916
## monthlyincome    8.36    2.79     3   268.7   1.5622  0.1989
## merged_score_zscore 0.01    0.01     1   269.8   0.0029  0.9574
## Stimuli_Type    885.59  885.59     1 13112.0 496.4617 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Random Effects for Attractiveness Model

```
## Random Effects Variance for Comprehensive Attractiveness Model:
```

```
## Groups      Name          Std.Dev.
## code        (Intercept) 1.31080141
## Model:block (Intercept) 0.77788977
## block        (Intercept) 0.00017621
## Residual                    1.33559097
```

5. Comprehensive Wealth Model

A comprehensive linear mixed-effects model for wealth was constructed, including all potential variables:

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Rating ~ Condition + sex + monthlyincome + merged_score_zscore +
## Stimuli_Type + (1 | block/Model) + (1 | code)
## Data: body_df_wealth
##
## REML criterion at convergence: 43764.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.5626 -0.5691 -0.0143  0.5608  4.8053
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## code        (Intercept) 0.8441    0.9188
## Model:block (Intercept) 0.1898    0.4356
## block        (Intercept) 0.0000    0.0000
## Residual                    1.4083    1.1867
## Number of obs: 13407, groups: code, 275; Model:block, 54; block, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    4.720e+00  1.519e-01  3.232e+02  31.078 < 2e-16 ***
## ConditionEconomy Flat  2.042e-01  3.239e-02  1.308e+04   6.303 3.01e-10 ***
## ConditionEconomy Heel  7.454e-01  3.241e-02  1.308e+04  22.998 < 2e-16 ***
## ConditionLuxury Flat  3.641e-01  3.239e-02  1.308e+04  11.240 < 2e-16 ***
## ConditionLuxury Heel  6.341e-01  3.242e-02  1.308e+04  19.559 < 2e-16 ***
## sexMale        -1.447e-01  1.297e-01  2.669e+02  -1.115  0.266
## monthlyincome.L  1.675e-01  2.948e-01  2.720e+02   0.568  0.570
## monthlyincome.Q  7.881e-02  2.523e-01  2.704e+02   0.312  0.755
## monthlyincome.C  2.116e-01  2.019e-01  2.723e+02   1.048  0.296
## merged_score_zscore -6.617e-02  7.105e-02  2.712e+02  -0.931  0.353
## Stimuli_TypeVideo -1.748e-01  2.050e-02  1.308e+04  -8.528 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) CndtEF CndtEH CndtLF CndtLH sexMal mnth.L mnth.Q mnth.C
## CndtnEcnmyF -0.107
## CndtnEcnmyH -0.107  0.500
## CndtnLxryFl -0.107  0.500  0.500
## CndtnLxryHl -0.106  0.500  0.499  0.500
## sexMale     -0.378  0.000  0.000  0.000  0.000
## mnthlyncm.L  0.714  0.000  0.000  0.000  0.000 -0.041
## mnthlyncm.Q  0.350  0.000  0.000  0.000  0.000  0.087  0.746
## mnthlyncm.C  0.136  0.000  0.000  0.000  0.000  0.050  0.296  0.602
## mrgd_scr_zs -0.115  0.000  0.000  0.000  0.000  0.375  0.011  0.010 -0.058
## Stiml_TypVd -0.068  0.000  0.000  0.000 -0.001  0.000  0.000  0.000  0.000
## mrgd__
## CndtnEcnmyF
## CndtnEcnmyH
## CndtnLxryFl
## CndtnLxryHl
## sexMale
## mnthlyncm.L

```

```
## mnthlyncm.Q
## mnthlyncm.C
## mrgd_scr_zs
## Stiml_TypVd 0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

ANOVA for Fixed Effects in Wealth Model

```
## Type III Analysis of Variance Table with Satterthwaite's method
##
##              Sum Sq Mean Sq NumDF    DenDF  F value Pr(>F)
## Condition      1000.40  250.100     4  13084.1  177.5854 <2e-16 ***
## sex              1.75    1.751     1    266.9   1.2436 0.2658
## monthlyincome    2.78    0.927     3    269.9   0.6582 0.5784
## merged_score_zscore 1.22    1.221     1    271.2   0.8671 0.3526
## Stimuli_Type     102.41  102.415     1  13084.1  72.7205 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Random Effects for Wealth Model

```
## Random Effects Variance for Comprehensive Wealth Model:
```

```
## Groups      Name          Std.Dev.
## code        (Intercept) 0.91877
## Model:block (Intercept) 0.43565
## block        (Intercept) 0.00000
## Residual                    1.18673
```

6. Model Comparison: Testing Variable Contributions

To assess the contribution of each additional variable, stepwise comparisons were conducted by removing one variable at a time:

6.1 Testing Sex Contribution

```
## Sex contribution to Attractiveness model:
```

```
## Data: body_df_attract
## Models:
## model_attract_no_sex: Rating ~ Condition + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)
## model_attract_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)
##
##              npar    AIC    BIC loglik -2*log(L)  Chisq Df Pr(>Chisq)
## model_attract_no_sex   14 47187 47292 -23579      47159
## model_attract_comp     15 47188 47300 -23579      47158 0.7497  1      0.3866
```

```
##  
## Sex contribution to Wealth model:
```

```
## Data: body_df_wealth  
## Models:  
## model_wealth_no_sex: Rating ~ Condition + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)  
## model_wealth_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)  
##  
##           npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)  
## model_wealth_no_sex    14 43753 43858 -21863      43725  
## model_wealth_comp     15 43754 43867 -21862      43724 1.2662  1      0.2605
```

6.2 Testing Monthly Income Contribution

```
## Monthly Income contribution to Attractiveness model:
```

```
## Data: body_df_attract  
## Models:  
## model_attract_no_income: Rating ~ Condition + sex + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)  
## model_attract_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)  
##  
##           npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)  
## model_attract_no_income    12 47187 47277 -23581      47163  
## model_attract_comp        15 47188 47300 -23579      47158 4.7415  3      0.1917
```

```
##  
## Monthly Income contribution to Wealth model:
```

```
## Data: body_df_wealth  
## Models:  
## model_wealth_no_income: Rating ~ Condition + sex + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)  
## model_wealth_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)  
##  
##           npar    AIC    BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)  
## model_wealth_no_income    12 43750 43840 -21863      43726  
## model_wealth_comp        15 43754 43867 -21862      43724 2.0067  3      0.571
```

6.3 Testing Brand Knowledge Score Contribution

```
## Brand Knowledge Score contribution to Attractiveness model:
```

```
## Data: body_df_attract
## Models:
## model_attract_no_brand: Rating ~ Condition + sex + monthlyincome + Stimuli_Type + (1 | block/Model) + (1 | code)
## model_attract_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)
##
##          npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_attract_no_brand    14 47186 47291 -23579      47158
## model_attract_comp        15 47188 47300 -23579      47158 0.0029  1      0.9568
```

```
##
## Brand Knowledge Score contribution to Wealth model:
```

```
## Data: body_df_wealth
## Models:
## model_wealth_no_brand: Rating ~ Condition + sex + monthlyincome + Stimuli_Type + (1 | block/Model) + (1 | code)
## model_wealth_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)
##
##          npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_wealth_no_brand    14 43753 43858 -21863      43725
## model_wealth_comp        15 43754 43867 -21862      43724 0.884  1      0.3471
```

6.4 Testing Stimuli Type Contribution

```
## Stimuli Type contribution to Attractiveness model:
```

```
## Data: body_df_attract
## Models:
## model_attract_no_stimuli: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + (1 | block/Model) + (1 | code)
## model_attract_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimuli_Type + (1 | block/Model) + (1 | code)
##
##          npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_attract_no_stimuli    14 47673 47778 -23823      47645
## model_attract_comp        15 47188 47300 -23579      47158 487.47  1 < 2.2e-16
##
## model_attract_no_stimuli
## model_attract_comp      ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Stimuli Type contribution to Wealth model:
```



```
## Data: body_df_wealth
## Models:
## model_wealth_no_stimuli: Rating ~ Condition + sex + monthlyincome + merged_score_zscore +
(1 | block/Model) + (1 | code)
## model_wealth_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimul
i_Type + (1 | block/Model) + (1 | code)
##
##          npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_wealth_no_stimuli    14 43825 43930 -21898      43797
## model_wealth_comp          15 43754 43867 -21862      43724 72.546   1 < 2.2e-16
##
## model_wealth_no_stimuli
## model_wealth_comp      ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6.5 Testing Model (nested in block) Contribution

```
## Nested Model contribution to Attractiveness model:
```

```
## Data: body_df_attract
## Models:
## model_attract_no_nested: Rating ~ Condition + sex + monthlyincome + merged_score_zscore +
Stimuli_Type + (1 | block) + (1 | code)
## model_attract_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimu
li_Type + (1 | block/Model) + (1 | code)
##
##          npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_attract_no_nested    14 50473 50578 -25223      50445
## model_attract_comp          15 47188 47300 -23579      47158 3287.1   1 < 2.2e-16
##
## model_attract_no_nested
## model_attract_comp      ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Nested Model contribution to Wealth model:
```

```
## Data: body_df_wealth
## Models:
## model_wealth_no_nested: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + S
timuli_Type + (1 | block) + (1 | code)
## model_wealth_comp: Rating ~ Condition + sex + monthlyincome + merged_score_zscore + Stimul
i_Type + (1 | block/Model) + (1 | code)
##
##          npar   AIC   BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## model_wealth_no_nested    14 45062 45167 -22517      45034
## model_wealth_comp          15 43754 43867 -21862      43724 1309.5   1 < 2.2e-16
##
## model_wealth_no_nested
## model_wealth_comp      ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

7. AIC and BIC Comparison for Model Selection

To further evaluate model selection, AIC and BIC values were compared between the comprehensive and simplified models:

```
## AIC and BIC comparison for Attractiveness models:
```

```
##           Model      AIC      BIC
## 1 Comprehensive 47222.76 47335.34
## 2           Simple 50862.96 50923.01
```

```
##
## AIC and BIC comparison for Wealth models:
```

```
##           Model      AIC      BIC
## 1 Comprehensive 43794.64 43907.19
## 2           Simple 45142.00 45202.03
```

PART II: Supplement B2: Unsplit models (without stimulus type separation)

Data Cleaning and Preparation

8. Attractiveness Model with Interaction

A linear mixed-effects model for attractiveness was constructed including the interaction between Condition and Stimuli_Type:

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Rating ~ Condition * Stimuli_Type + (1 | block/Model) + (1 |
##      code)
##      Data: body_df_attract
##
## REML criterion at convergence: 47670.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.1231 -0.6220 -0.0116  0.6116  4.3467
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
##      code      (Intercept) 1.7190   1.3111
##      Model:block (Intercept) 0.6136   0.7833
##      block      (Intercept) 0.0000   0.0000
##      Residual                1.7539   1.3244
## Number of obs: 13633, groups:  code, 279; Model:block, 54; block, 11
##
## Fixed effects:
##
##              Estimate Std. Error      df
## (Intercept)      5.170e+00  1.373e-01  1.257e+02
## ConditionEconomy Flat      -3.390e-02  5.074e-02  1.330e+04
## ConditionEconomy Heel       3.443e-01  5.071e-02  1.330e+04
## ConditionLuxury Flat       5.657e-02  5.070e-02  1.330e+04
## ConditionLuxury Heel       3.270e-01  5.075e-02  1.330e+04
## Stimuli_TypeVideo      -6.763e-02  5.070e-02  1.330e+04
## ConditionEconomy Flat:Stimuli_TypeVideo -2.302e-01  7.172e-02  1.330e+04
## ConditionEconomy Heel:Stimuli_TypeVideo -6.500e-01  7.172e-02  1.330e+04
## ConditionLuxury Flat:Stimuli_TypeVideo -3.716e-01  7.171e-02  1.330e+04
## ConditionLuxury Heel:Stimuli_TypeVideo -9.689e-01  7.175e-02  1.330e+04
##
##              t value Pr(>|t|)
## (Intercept)      37.658 < 2e-16 ***
## ConditionEconomy Flat      -0.668  0.50406
## ConditionEconomy Heel       6.789 1.18e-11 ***
## ConditionLuxury Flat       1.116  0.26460
## ConditionLuxury Heel       6.443 1.21e-10 ***
## Stimuli_TypeVideo      -1.334  0.18228
## ConditionEconomy Flat:Stimuli_TypeVideo -3.210  0.00133 **
## ConditionEconomy Heel:Stimuli_TypeVideo -9.062 < 2e-16 ***
## ConditionLuxury Flat:Stimuli_TypeVideo -5.182 2.23e-07 ***
## ConditionLuxury Heel:Stimuli_TypeVideo -13.503 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) CndtEF CndtEH CndtLF CndtLH Stm_TV CEF:S_ CEH:S_ CLF:S_
## CndtnEcnmyF -0.185
## CndtnEcnmyH -0.185  0.500
## CndtnLxryF1 -0.185  0.500  0.500
## CndtnLxryH1 -0.185  0.499  0.500  0.500
## Stiml_TypVd -0.185  0.500  0.500  0.500  0.500
## CndtEF:S_TV  0.131 -0.707 -0.354 -0.354 -0.353 -0.707
## CndtEH:S_TV  0.131 -0.353 -0.707 -0.354 -0.353 -0.707  0.500

```

```
## CndtLF:S_TV  0.131 -0.353 -0.354 -0.707 -0.353 -0.707  0.500  0.500
## CndtLH:S_TV  0.131 -0.353 -0.353 -0.353 -0.707 -0.707  0.500  0.500  0.500
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

ANOVA for Fixed Effects in Attractiveness Model with Interaction

```
## Type III Analysis of Variance Table with Satterthwaite's method
##
##              Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## Condition           80.14   20.03     4 13302  11.423 2.959e-09 ***
## Stimuli_Type        892.56  892.56     1 13302 508.895 < 2.2e-16 ***
## Condition:Stimuli_Type 385.58   96.39     4 13302  54.960 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means for Condition by Stimuli_Type in Attractiveness Model

```
## Stimuli_Type = Image:
## Condition    emmean    SE df asymp.LCL asymp.UCL
## Control      5.17 0.137 Inf      4.90      5.44
## Economy Flat  5.14 0.137 Inf      4.87      5.41
## Economy Heel  5.51 0.137 Inf      5.25      5.78
## Luxury Flat   5.23 0.137 Inf      4.96      5.50
## Luxury Heel   5.50 0.137 Inf      5.23      5.77
##
## Stimuli_Type = Video:
## Condition    emmean    SE df asymp.LCL asymp.UCL
## Control      5.10 0.137 Inf      4.83      5.37
## Economy Flat  4.84 0.137 Inf      4.57      5.11
## Economy Heel  4.80 0.137 Inf      4.53      5.07
## Luxury Flat   4.79 0.137 Inf      4.52      5.06
## Luxury Heel   4.46 0.137 Inf      4.19      4.73
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Calculate R-squared values for the attractiveness model

```
##              R2m      R2c
## [1,] 0.02379299 0.5810165
```

```
## Attractiveness Model R-squared:
```

```
## R2m (Fixed effects only): 0.0238
```

```
## R2c (Fixed + Random effects): 0.581
```

```
## Random effects contribution: 0.5572
```

Visualizing Attractiveness Results by Condition and Stimuli_Type

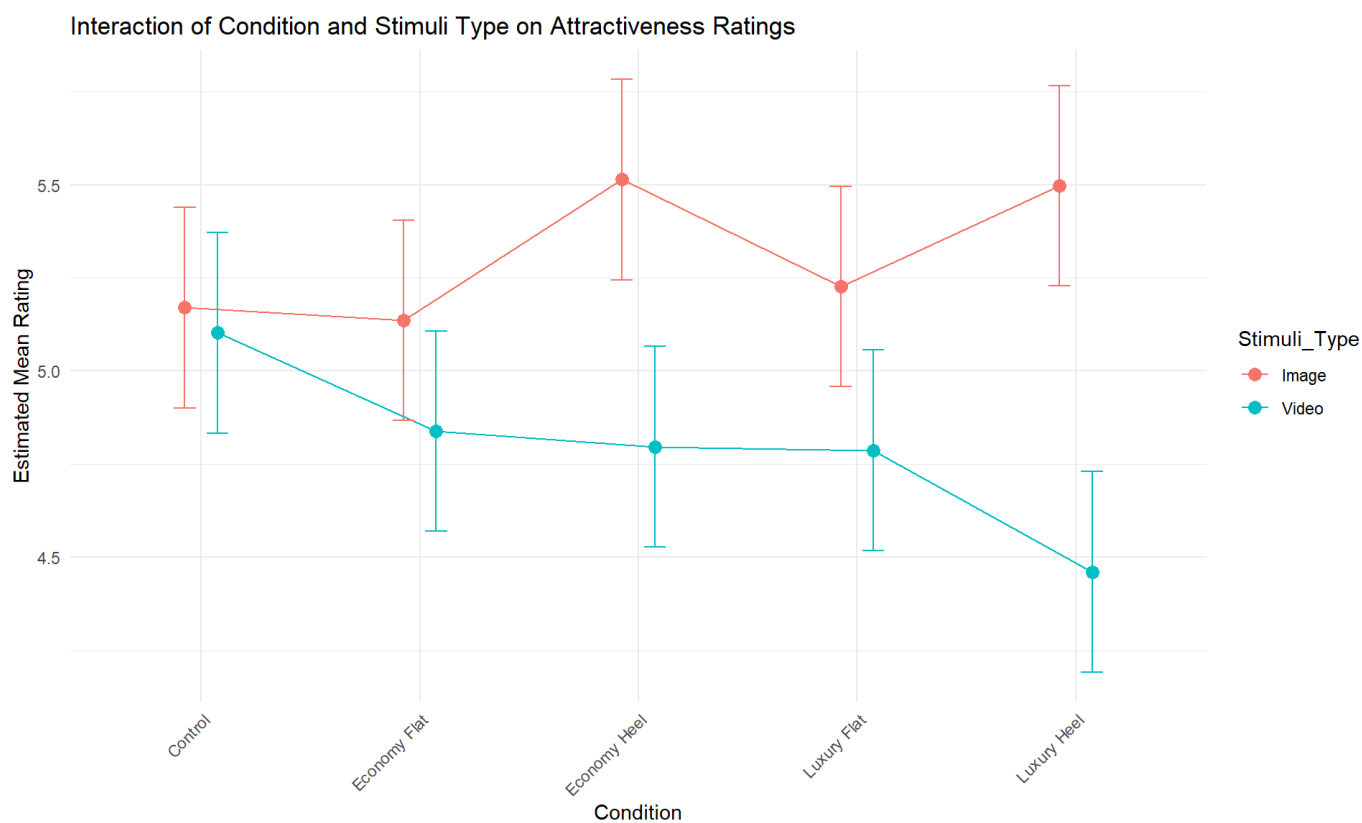


Figure B2.1. Interaction between footwear condition and stimulus type (image vs. video) on attractiveness ratings. Error bars represent 95% confidence intervals.

Pairwise Comparisons for Condition by Stimuli_Type in

Attractiveness Model

```
## Stimuli_Type = Image:
## contrast      estimate      SE  df z.ratio p.value
## Control - Economy Flat      0.03390 0.0507 Inf   0.668  0.9632
## Control - Economy Heel     -0.34427 0.0507 Inf  -6.789 <.0001
## Control - Luxury Flat      -0.05657 0.0507 Inf  -1.116  0.7984
## Control - Luxury Heel      -0.32699 0.0508 Inf  -6.443 <.0001
## Economy Flat - Economy Heel -0.37818 0.0507 Inf  -7.453 <.0001
## Economy Flat - Luxury Flat  -0.09047 0.0507 Inf  -1.783  0.3833
## Economy Flat - Luxury Heel  -0.36089 0.0508 Inf  -7.107 <.0001
## Economy Heel - Luxury Flat   0.28771 0.0507 Inf   5.674 <.0001
## Economy Heel - Luxury Heel   0.01729 0.0508 Inf   0.341  0.9971
## Luxury Flat - Luxury Heel   -0.27042 0.0507 Inf  -5.329 <.0001
##
## Stimuli_Type = Video:
## contrast      estimate      SE  df z.ratio p.value
## Control - Economy Flat      0.26415 0.0507 Inf   5.211 <.0001
## Control - Economy Heel      0.30567 0.0507 Inf   6.027 <.0001
## Control - Luxury Flat       0.31504 0.0507 Inf   6.212 <.0001
## Control - Luxury Heel       0.64189 0.0507 Inf  12.655 <.0001
## Economy Flat - Economy Heel  0.04153 0.0507 Inf   0.819  0.9249
## Economy Flat - Luxury Flat   0.05089 0.0507 Inf   1.004  0.8538
## Economy Flat - Luxury Heel   0.37774 0.0507 Inf   7.448 <.0001
## Economy Heel - Luxury Flat   0.00936 0.0507 Inf   0.185  0.9997
## Economy Heel - Luxury Heel   0.33622 0.0507 Inf   6.626 <.0001
## Luxury Flat - Luxury Heel    0.32685 0.0507 Inf   6.441 <.0001
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 5 estimates
```

9. Wealth Model with Interaction

A linear mixed-effects model for wealth was constructed including the interaction between Condition and Stimuli_Type:

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Rating ~ Condition * Stimuli_Type + (1 | block/Model) + (1 |
## code)
## Data: body_df_wealth
##
## REML criterion at convergence: 44395.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.5286 -0.5751 -0.0159  0.5588  4.7829
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## code        (Intercept) 0.8498    0.9218
## Model:block (Intercept) 0.1943    0.4408
## block        (Intercept) 0.0000    0.0000
## Residual                    1.4066    1.1860
## Number of obs: 13603, groups: code, 279; Model:block, 54; block, 11
##
## Fixed effects:
##
##              Estimate Std. Error      df
## (Intercept)    4.534e+00  8.771e-02  1.841e+02
## ConditionEconomy Flat    1.946e-01  4.548e-02  1.327e+04
## ConditionEconomy Heel    7.817e-01  4.548e-02  1.327e+04
## ConditionLuxury Flat    3.798e-01  4.546e-02  1.327e+04
## ConditionLuxury Heel    7.965e-01  4.554e-02  1.327e+04
## Stimuli_TypeVideo    -9.442e-02  4.545e-02  1.327e+04
## ConditionEconomy Flat:Stimuli_TypeVideo    2.230e-02  6.428e-02  1.327e+04
## ConditionEconomy Heel:Stimuli_TypeVideo    -7.445e-02  6.431e-02  1.327e+04
## ConditionLuxury Flat:Stimuli_TypeVideo    -2.573e-02  6.428e-02  1.327e+04
## ConditionLuxury Heel:Stimuli_TypeVideo    -3.298e-01  6.434e-02  1.327e+04
##
##              t value Pr(>|t|)
## (Intercept)    51.690 < 2e-16 ***
## ConditionEconomy Flat    4.278 1.90e-05 ***
## ConditionEconomy Heel    17.186 < 2e-16 ***
## ConditionLuxury Flat    8.355 < 2e-16 ***
## ConditionLuxury Heel    17.492 < 2e-16 ***
## Stimuli_TypeVideo    -2.077 0.0378 *
## ConditionEconomy Flat:Stimuli_TypeVideo    0.347 0.7287
## ConditionEconomy Heel:Stimuli_TypeVideo    -1.158 0.2470
## ConditionLuxury Flat:Stimuli_TypeVideo    -0.400 0.6890
## ConditionLuxury Heel:Stimuli_TypeVideo    -5.127 2.99e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) CndtEF CndtEH CndtLF CndtLH Stm_TV CEF:S_ CEH:S_ CLF:S_
## CndtnEcnmyF -0.259
## CndtnEcnmyH -0.259 0.500
## CndtnLxryF1 -0.259 0.500 0.500
## CndtnLxryH1 -0.259 0.499 0.499 0.499
## Stiml_TypVd -0.259 0.500 0.500 0.500 0.499
## CndtEF:S_TV 0.183 -0.708 -0.354 -0.354 -0.353 -0.707
## CndtEH:S_TV 0.183 -0.353 -0.707 -0.354 -0.353 -0.707 0.500

```

```
## CndtLF:S_TV  0.183 -0.354 -0.354 -0.707 -0.353 -0.707  0.500  0.500
## CndtLH:S_TV  0.183 -0.353 -0.353 -0.353 -0.708 -0.706  0.499  0.499  0.500
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

ANOVA for Fixed Effects in Wealth Model with Interaction

```
## Type III Analysis of Variance Table with Satterthwaite's method
##
##              Sum Sq Mean Sq NumDF DenDF  F value    Pr(>F)
## Condition      1007.95   251.99     4 13272 179.1457 < 2.2e-16 ***
## Stimuli_Type     105.28   105.28     1 13272  74.8453 < 2.2e-16 ***
## Condition:Stimuli_Type    55.84    13.96     4 13272   9.9248 5.131e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means for Condition by Stimuli_Type in Wealth Model

```
## Stimuli_Type = Image:
## Condition      emmean      SE df asymp.LCL asymp.UCL
## Control         4.53 0.0877 Inf      4.36      4.71
## Economy Flat     4.73 0.0877 Inf      4.56      4.90
## Economy Heel     5.32 0.0877 Inf      5.14      5.49
## Luxury Flat      4.91 0.0877 Inf      4.74      5.09
## Luxury Heel      5.33 0.0877 Inf      5.16      5.50
##
## Stimuli_Type = Video:
## Condition      emmean      SE df asymp.LCL asymp.UCL
## Control         4.44 0.0877 Inf      4.27      4.61
## Economy Flat     4.66 0.0877 Inf      4.48      4.83
## Economy Heel     5.15 0.0877 Inf      4.97      5.32
## Luxury Flat      4.79 0.0877 Inf      4.62      4.97
## Luxury Heel      4.91 0.0877 Inf      4.73      5.08
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Calculate R-squared values for the wealth model

```
##              R2m      R2c
## [1,] 0.03386218 0.4454785
```

```
## Wealth Model R-squared:
```

```
## R2m (Fixed effects only): 0.0339
```

```
## R2c (Fixed + Random effects): 0.4455
```

```
## Random effects contribution: 0.4116
```


Visualizing Wealth Results by Condition and Stimuli_Type

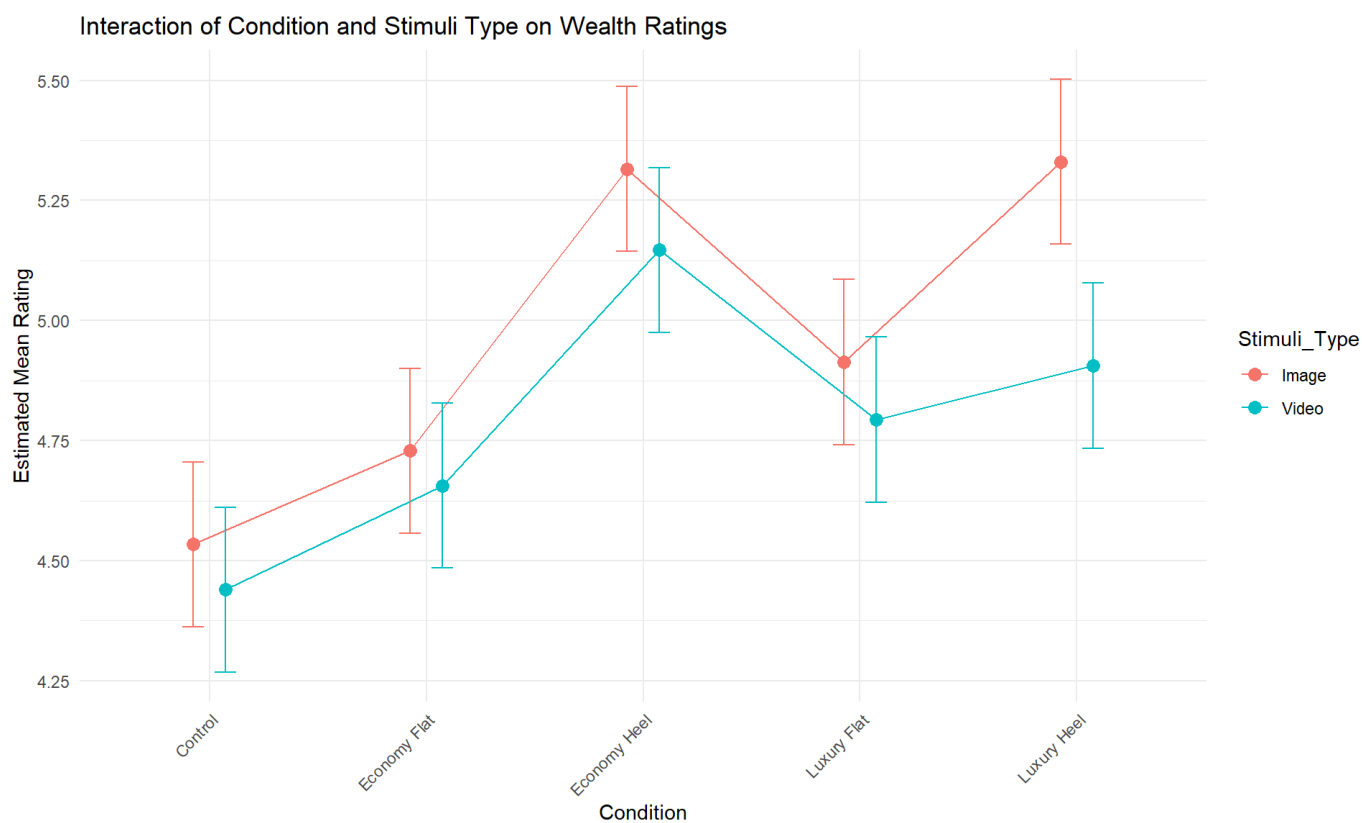


Figure B2.2. Interaction between footwear condition and stimulus type (image vs. video) on wealth perception ratings. Error bars represent 95% confidence intervals.

Pairwise Comparisons for Condition by Stimuli_Type in Wealth

Model

```
## Stimuli_Type = Image:
## contrast estimate SE df z.ratio p.value
## Control - Economy Flat -0.1946 0.0455 Inf -4.278 0.0002
## Control - Economy Heel -0.7817 0.0455 Inf -17.186 <.0001
## Control - Luxury Flat -0.3798 0.0455 Inf -8.355 <.0001
## Control - Luxury Heel -0.7965 0.0455 Inf -17.492 <.0001
## Economy Flat - Economy Heel -0.5871 0.0455 Inf -12.903 <.0001
## Economy Flat - Luxury Flat -0.1852 0.0455 Inf -4.073 0.0004
## Economy Flat - Luxury Heel -0.6019 0.0456 Inf -13.214 <.0001
## Economy Heel - Luxury Flat 0.4019 0.0455 Inf 8.837 <.0001
## Economy Heel - Luxury Heel -0.0148 0.0456 Inf -0.325 0.9976
## Luxury Flat - Luxury Heel -0.4167 0.0455 Inf -9.153 <.0001
##
## Stimuli_Type = Video:
## contrast estimate SE df z.ratio p.value
## Control - Economy Flat -0.2169 0.0454 Inf -4.774 <.0001
## Control - Economy Heel -0.7072 0.0455 Inf -15.555 <.0001
## Control - Luxury Flat -0.3541 0.0454 Inf -7.792 <.0001
## Control - Luxury Heel -0.4667 0.0455 Inf -10.267 <.0001
## Economy Flat - Economy Heel -0.4904 0.0455 Inf -10.787 <.0001
## Economy Flat - Luxury Flat -0.1372 0.0454 Inf -3.020 0.0213
## Economy Flat - Luxury Heel -0.2498 0.0454 Inf -5.497 <.0001
## Economy Heel - Luxury Flat 0.3532 0.0455 Inf 7.766 <.0001
## Economy Heel - Luxury Heel 0.2406 0.0455 Inf 5.289 <.0001
## Luxury Flat - Luxury Heel -0.1126 0.0455 Inf -2.477 0.0958
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 5 estimates
```

10. Combined Plot: Condition × Dimension × Stimuli_Type

A combined plot was created to visualize the interaction between Condition, Dimension, and Stimuli_Type:

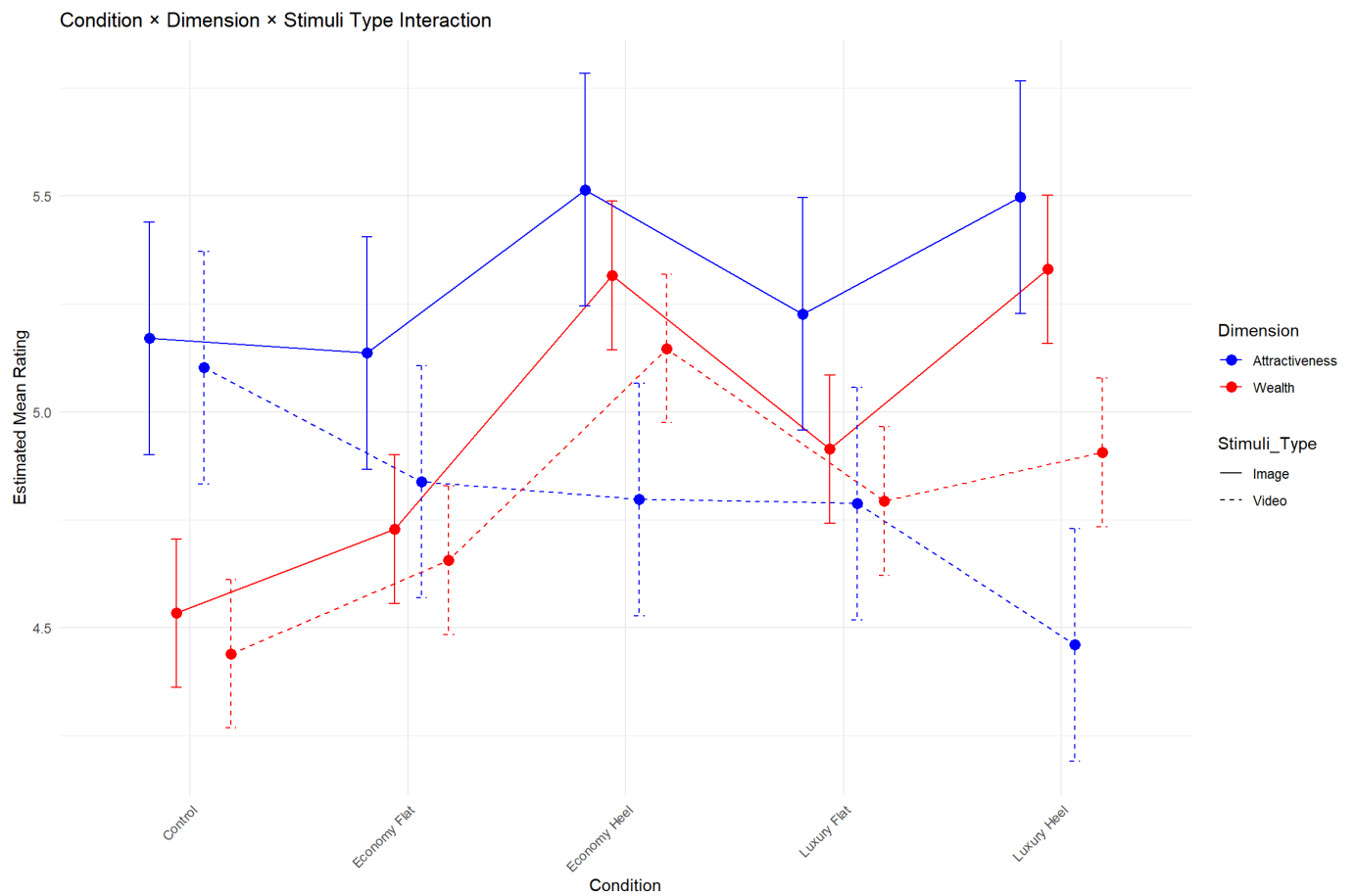


Figure B2.3. Three-way interaction between footwear condition, dimension (attractiveness vs. wealth), and stimulus type. Solid lines represent image stimuli; dashed lines represent video stimuli. Error bars represent 95% confidence intervals.

PART III: Supplement B3: Complete models including control condition (Split by StimuliType)

11. Models for Image Stimuli Only

Attractiveness Model for Image Stimuli

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Condition + (1 | block/Model) + (1 | code)  
## Data: body_df_attract_image  
##  
## REML criterion at convergence: 22528.6  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -5.1111 -0.5959 -0.0052  0.5790  4.6063  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept) 1.989      1.4103  
## Model:block (Intercept) 0.677      0.8228  
## block       (Intercept) 0.000      0.0000  
## Residual                    1.336      1.1560  
## Number of obs: 6814, groups: code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)      5.16903    0.14384 124.06307  35.935 < 2e-16 ***  
## ConditionEconomy Flat -0.03432    0.04429 6487.97626  -0.775  0.438  
## ConditionEconomy Heel  0.34415    0.04427 6487.98276   7.774 8.77e-15 ***  
## ConditionLuxury Flat   0.05591    0.04426 6487.96655   1.263  0.207  
## ConditionLuxury Heel   0.32696    0.04430 6487.99434   7.380 1.78e-13 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) CndtEF CndtEH CndtLF  
## CndtnEcnmyF -0.154  
## CndtnEcnmyH -0.154  0.500  
## CndtnLxryFl -0.154  0.500  0.500  
## CndtnLxryHl -0.154  0.499  0.500  0.500  
## optimizer (nloptwrap) convergence code: 0 (OK)  
## boundary (singular) fit: see help('isSingular')
```

ANOVA for Fixed Effects in Attractiveness Model (Image Stimuli)

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## Condition 182.09  45.522     4   6488  34.065 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means for Condition in Attractiveness Model (Image Stimuli)

```
## Condition      emmean      SE df asymp.LCL asymp.UCL
## Control         5.17 0.144 Inf      4.89      5.45
## Economy Flat    5.13 0.144 Inf      4.85      5.42
## Economy Heel    5.51 0.144 Inf      5.23      5.80
## Luxury Flat     5.22 0.144 Inf      4.94      5.51
## Luxury Heel     5.50 0.144 Inf      5.21      5.78
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Calculate R-squared values for the attractiveness model

```
##           R2m      R2c
## [1,] 0.006634421 0.6683305
```

```
## Attractiveness Model R-squared:
```

```
## R²m (Fixed effects only): 0.0066
```

```
## R²c (Fixed + Random effects): 0.6683
```

```
## Random effects contribution: 0.6617
```

Visualizing Attractiveness Results for Image Stimuli

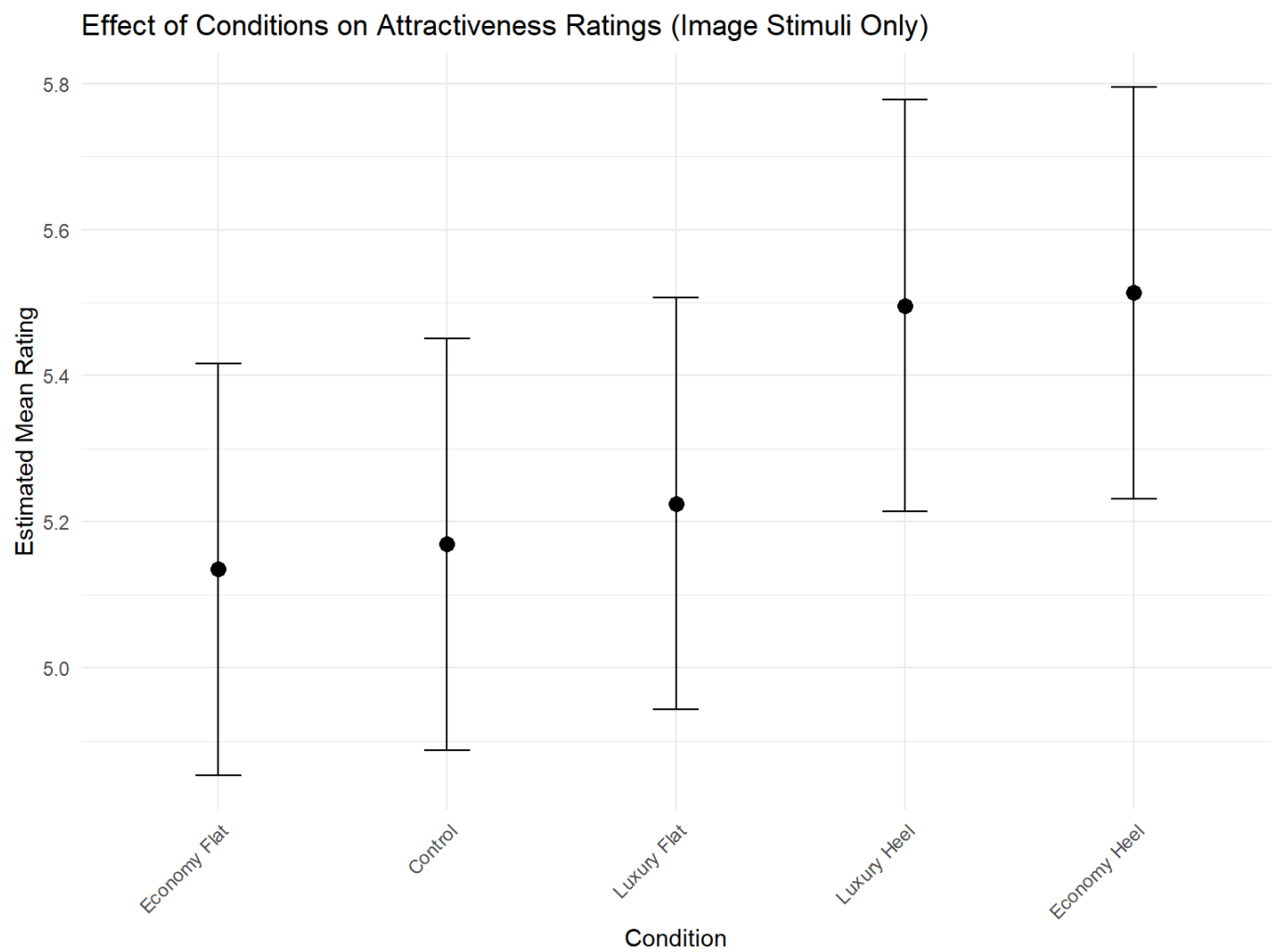


Figure B3.1. Effect of footwear conditions on attractiveness ratings for image stimuli, including control condition. Error bars represent 95% confidence intervals.

Pairwise Comparisons for Condition in Attractiveness Model (Image Stimuli)

##	contrast	estimate	SE	df	z.ratio	p.value
##	Control - Economy Flat	0.0343	0.0443	Inf	0.775	0.9379
##	Control - Economy Heel	-0.3442	0.0443	Inf	-7.774	<.0001
##	Control - Luxury Flat	-0.0559	0.0443	Inf	-1.263	0.7138
##	Control - Luxury Heel	-0.3270	0.0443	Inf	-7.380	<.0001
##	Economy Flat - Economy Heel	-0.3785	0.0443	Inf	-8.545	<.0001
##	Economy Flat - Luxury Flat	-0.0902	0.0443	Inf	-2.038	0.2480
##	Economy Flat - Luxury Heel	-0.3613	0.0443	Inf	-8.150	<.0001
##	Economy Heel - Luxury Flat	0.2882	0.0443	Inf	6.512	<.0001
##	Economy Heel - Luxury Heel	0.0172	0.0443	Inf	0.388	0.9952
##	Luxury Flat - Luxury Heel	-0.2711	0.0443	Inf	-6.119	<.0001
##						
##	Degrees-of-freedom method: asymptotic					
##	P value adjustment: tukey method for comparing a family of 5 estimates					

Wealth Model for Image Stimuli

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Condition + (1 | block/Model) + (1 | code)  
## Data: body_df_wealth_image  
##  
## REML criterion at convergence: 20811.7  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -5.0202 -0.5717 -0.0140  0.5552  4.4834  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept) 9.403e-01 9.697e-01  
## Model:block (Intercept) 1.935e-01 4.399e-01  
## block        (Intercept) 1.581e-09 3.976e-05  
## Residual                1.077e+00 1.038e+00  
## Number of obs: 6794, groups: code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)      4.532e+00  8.810e-02 1.798e+02  51.442 < 2e-16 ***  
## ConditionEconomy Flat 1.939e-01  3.980e-02 6.468e+03   4.872 1.13e-06 ***  
## ConditionEconomy Heel 7.818e-01  3.981e-02 6.468e+03  19.641 < 2e-16 ***  
## ConditionLuxury Flat  3.785e-01  3.978e-02 6.468e+03   9.515 < 2e-16 ***  
## ConditionLuxury Heel  7.979e-01  3.985e-02 6.468e+03  20.023 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) CndtEF CndtEH CndtLF  
## CndtnEcnmyF -0.226  
## CndtnEcnmyH -0.226  0.500  
## CndtnLxryFl -0.226  0.500  0.500  
## CndtnLxryHl -0.225  0.499  0.499  0.499  
## optimizer (nloptwrap) convergence code: 0 (OK)  
## boundary (singular) fit: see help('isSingular')
```

ANOVA for Fixed Effects in Wealth Model (Image Stimuli)

```
## Type III Analysis of Variance Table with Satterthwaite's method  
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)  
## Condition 682.14  170.54      4 6468.1  158.32 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means for Condition in Wealth Model (Image

Stimuli)

##	Condition	emmean	SE	df	asyp.LCL	asyp.UCL
##	Control	4.53	0.0881	Inf	4.36	4.70
##	Economy Flat	4.73	0.0881	Inf	4.55	4.90
##	Economy Heel	5.31	0.0881	Inf	5.14	5.49
##	Luxury Flat	4.91	0.0881	Inf	4.74	5.08
##	Luxury Heel	5.33	0.0881	Inf	5.16	5.50
##						
##	Degrees-of-freedom method: asymptotic					
##	Confidence level used: 0.95					

Calculate R-squared values for the wealth model

##	R2m	R2c
## [1,]	0.04345892	0.5339808

Wealth Model R-squared:

R²m (Fixed effects only): 0.0435

R²c (Fixed + Random effects): 0.534
--

Random effects contribution: 0.4905
--

Visualizing Wealth Results for Image Stimuli

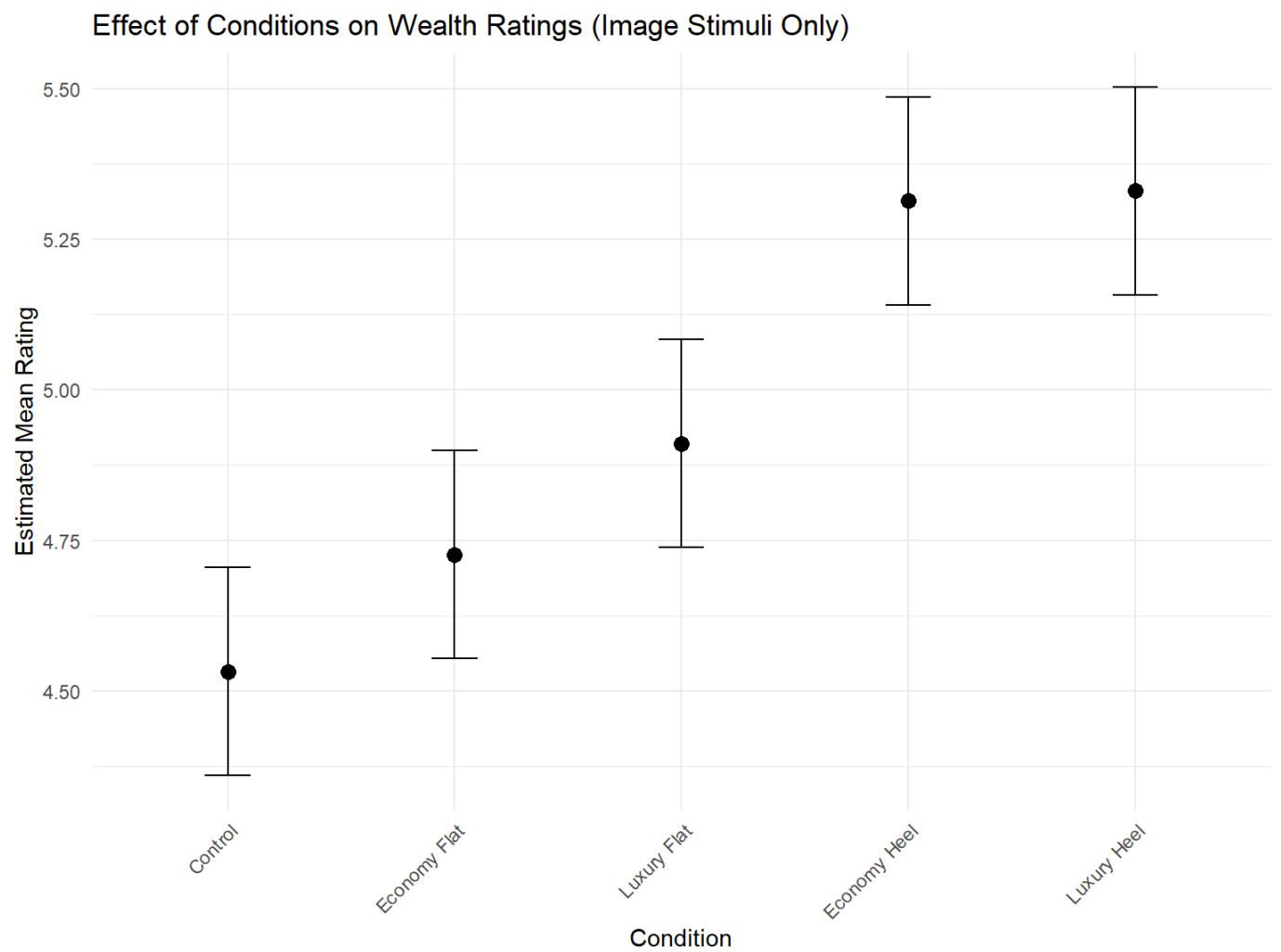


Figure B3.2. Effect of footwear conditions on wealth perception ratings for image stimuli, including control condition. Error bars represent 95% confidence intervals.

Pairwise Comparisons for Condition in Wealth Model (Image Stimuli)

##	contrast	estimate	SE	df	z.ratio	p.value
##	Control - Economy Flat	-0.1939	0.0398	Inf	-4.872	<.0001
##	Control - Economy Heel	-0.7818	0.0398	Inf	-19.641	<.0001
##	Control - Luxury Flat	-0.3785	0.0398	Inf	-9.515	<.0001
##	Control - Luxury Heel	-0.7979	0.0399	Inf	-20.023	<.0001
##	Economy Flat - Economy Heel	-0.5879	0.0398	Inf	-14.763	<.0001
##	Economy Flat - Luxury Flat	-0.1846	0.0398	Inf	-4.638	<.0001
##	Economy Flat - Luxury Heel	-0.6040	0.0399	Inf	-15.151	<.0001
##	Economy Heel - Luxury Flat	0.4033	0.0398	Inf	10.133	<.0001
##	Economy Heel - Luxury Heel	-0.0161	0.0399	Inf	-0.404	0.9944
##	Luxury Flat - Luxury Heel	-0.4194	0.0398	Inf	-10.526	<.0001
##						
##	Degrees-of-freedom method: asymptotic					
##	P value adjustment: tukey method for comparing a family of 5 estimates					

Comparison Between Dimensions for Image Stimuli

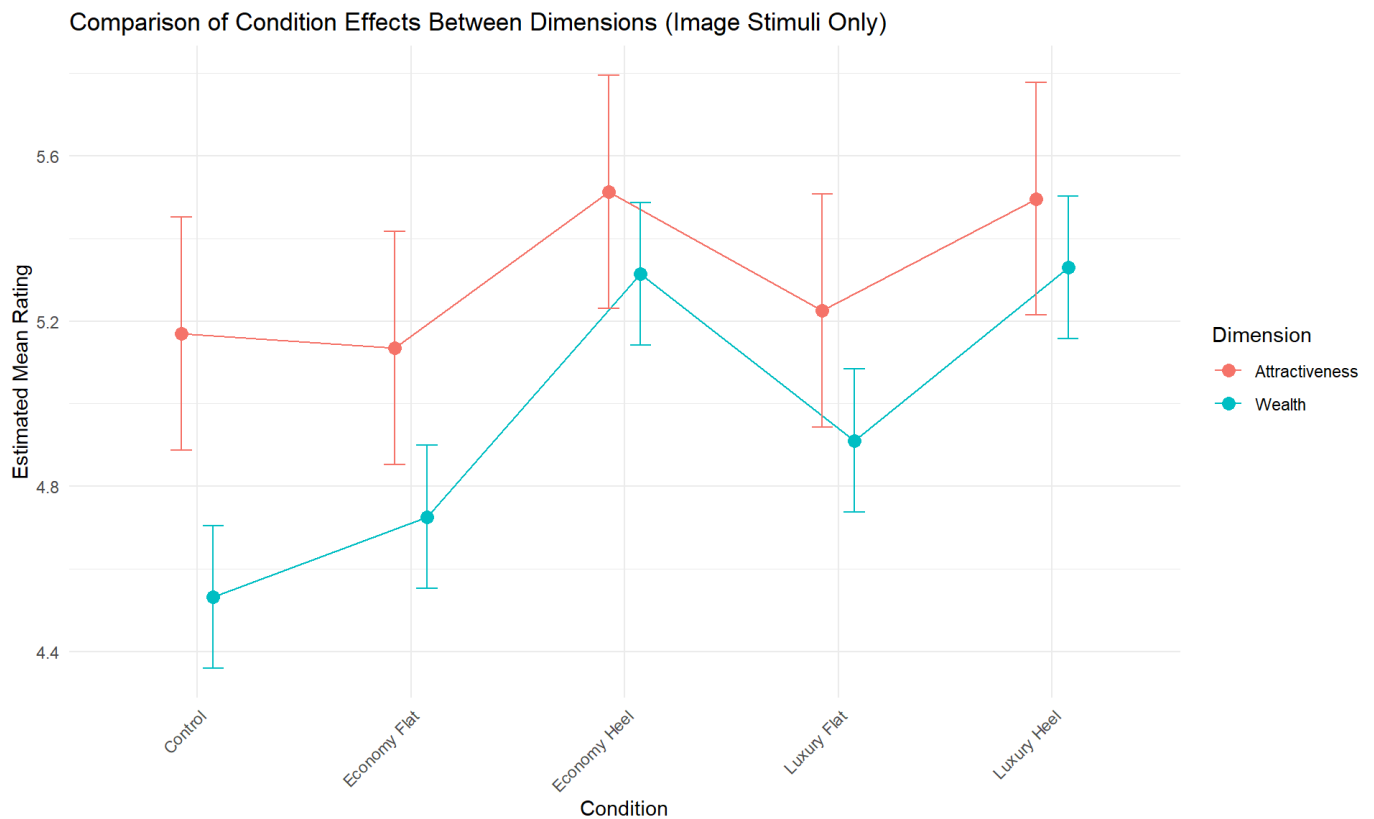


Figure B3.3. Comparison of footwear condition effects between attractiveness and wealth dimensions for image stimuli. Error bars represent 95% confidence intervals.

12. Models for Video Stimuli Only

Next, the data was filtered to include only Video stimuli:

Attractiveness Model for Video Stimuli

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Condition + (1 | block/Model) + (1 | code)  
## Data: body_df_attract_video  
##  
## REML criterion at convergence: 24118.3  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -5.1941 -0.6264 -0.0367  0.6031  4.0194  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept) 1.7047    1.3057  
## Model:block (Intercept) 0.7812    0.8838  
## block       (Intercept) 0.0000    0.0000  
## Residual                    1.7111    1.3081  
## Number of obs: 6819, groups: code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)      5.10366    0.14789  105.10690  34.510 < 2e-16 ***  
## ConditionEconomy Flat -0.26387    0.05006  6493.10837  -5.271 1.40e-07 ***  
## ConditionEconomy Heel -0.30457    0.05009  6493.13615  -6.080 1.27e-09 ***  
## ConditionLuxury Flat -0.31447    0.05009  6493.11634  -6.278 3.65e-10 ***  
## ConditionLuxury Heel -0.64010    0.05010  6493.16374 -12.776 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) CndtEF CndtEH CndtLF  
## CndtnEcnmyF -0.169  
## CndtnEcnmyH -0.169  0.500  
## CndtnLxryFl -0.169  0.500  0.500  
## CndtnLxryHl -0.169  0.500  0.500  0.500  
## optimizer (nloptwrap) convergence code: 0 (OK)  
## boundary (singular) fit: see help('isSingular')
```

ANOVA for Fixed Effects in Attractiveness Model (Video Stimuli)

```
## Type III Analysis of Variance Table with Satterthwaite's method  
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)  
## Condition  282.3   70.576      4 6493.1  41.246 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means for Condition in Attractiveness Model

(Video Stimuli)

##	Condition	emmean	SE	df	asympt.LCL	asympt.UCL
##	Control	5.10	0.148	Inf	4.81	5.39
##	Economy Flat	4.84	0.148	Inf	4.55	5.13
##	Economy Heel	4.80	0.148	Inf	4.51	5.09
##	Luxury Flat	4.79	0.148	Inf	4.50	5.08
##	Luxury Heel	4.46	0.148	Inf	4.17	4.75
##						
##	Degrees-of-freedom method: asymptotic					
##	Confidence level used: 0.95					

Calculate R-squared values for the attractiveness model

##	R2m	R2c
## [1,]	0.009771014	0.5962899

Attractiveness Model R-squared:

R²m (Fixed effects only): 0.0098

R²c (Fixed + Random effects): 0.5963

Random effects contribution: 0.5865

Visualizing Attractiveness Results for Video Stimuli

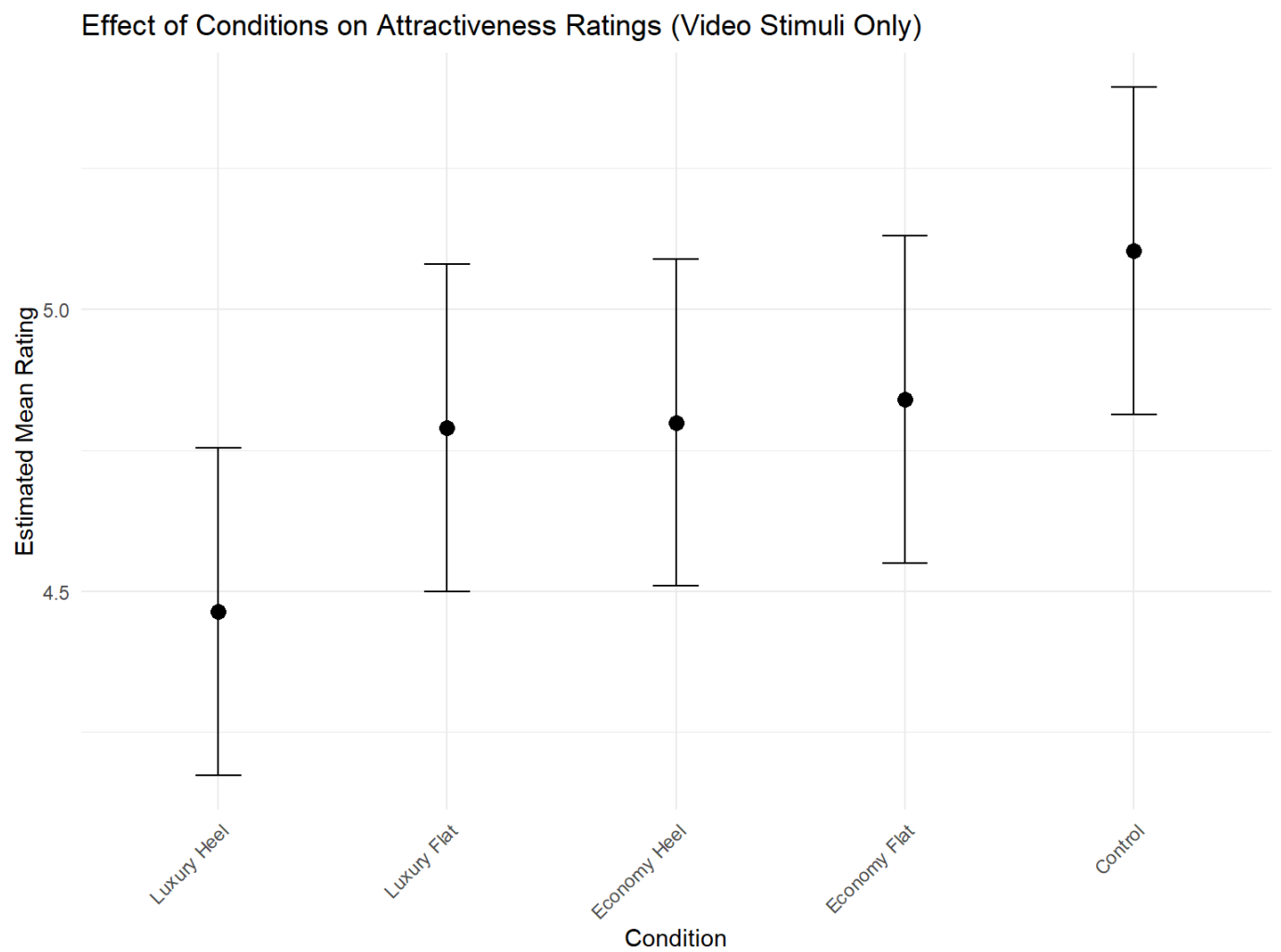


Figure B3.4. Effect of footwear conditions on attractiveness ratings for video stimuli, including control condition. Error bars represent 95% confidence intervals.

Pairwise Comparisons for Condition in Attractiveness Model (Video Stimuli)

##	contrast	estimate	SE	df	z.ratio	p.value
##	Control - Economy Flat	0.2639	0.0501	Inf	5.271	<.0001
##	Control - Economy Heel	0.3046	0.0501	Inf	6.080	<.0001
##	Control - Luxury Flat	0.3145	0.0501	Inf	6.278	<.0001
##	Control - Luxury Heel	0.6401	0.0501	Inf	12.776	<.0001
##	Economy Flat - Economy Heel	0.0407	0.0501	Inf	0.813	0.9269
##	Economy Flat - Luxury Flat	0.0506	0.0501	Inf	1.010	0.8508
##	Economy Flat - Luxury Heel	0.3762	0.0501	Inf	7.511	<.0001
##	Economy Heel - Luxury Flat	0.0099	0.0501	Inf	0.198	0.9997
##	Economy Heel - Luxury Heel	0.3355	0.0501	Inf	6.695	<.0001
##	Luxury Flat - Luxury Heel	0.3256	0.0501	Inf	6.497	<.0001
##						
##	Degrees-of-freedom method: asymptotic					
##	P value adjustment: tukey method for comparing a family of 5 estimates					

Wealth Model for Video Stimuli

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Condition + (1 | block/Model) + (1 | code)  
## Data: body_df_wealth_video  
##  
## REML criterion at convergence: 23072.6  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -4.1828 -0.5862 -0.0153  0.5623  4.9834  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept) 8.905e-01 0.9436448  
## Model:block (Intercept) 2.990e-01 0.5468359  
## block       (Intercept) 2.930e-08 0.0001712  
## Residual                    1.513e+00 1.2301948  
## Number of obs: 6809, groups: code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)    4.441e+00  9.930e-02 1.345e+02  44.725 < 2e-16 ***  
## ConditionEconomy Flat 2.167e-01  4.712e-02 6.483e+03   4.600 4.31e-06 ***  
## ConditionEconomy Heel 7.067e-01  4.716e-02 6.483e+03  14.984 < 2e-16 ***  
## ConditionLuxury Flat  3.533e-01  4.714e-02 6.483e+03   7.495 7.52e-14 ***  
## ConditionLuxury Heel  4.671e-01  4.715e-02 6.483e+03   9.906 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) CndtEF CndtEH CndtLF  
## CndtnEcnmyF -0.237  
## CndtnEcnmyH -0.237  0.500  
## CndtnLxryFl -0.237  0.500  0.500  
## CndtnLxryHl -0.237  0.500  0.499  0.500
```

ANOVA for Fixed Effects in Wealth Model (Video Stimuli)

```
## Type III Analysis of Variance Table with Satterthwaite's method  
##      Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)  
## Condition 382.66  95.666      4 6483.3  63.213 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means for Condition in Wealth Model (Video

Stimuli)

##	Condition	emmean	SE	df	asyp.LCL	asyp.UCL
##	Control	4.44	0.0993	Inf	4.25	4.64
##	Economy Flat	4.66	0.0993	Inf	4.46	4.85
##	Economy Heel	5.15	0.0993	Inf	4.95	5.34
##	Luxury Flat	4.79	0.0993	Inf	4.60	4.99
##	Luxury Heel	4.91	0.0993	Inf	4.71	5.10
##						
##	Degrees-of-freedom method: asymptotic					
##	Confidence level used: 0.95					

Calculate R-squared values for the wealth model

##	R2m	R2c
## [1,]	0.0203768	0.4514945

Wealth Model R-squared:

R ² m (Fixed effects only): 0.0204
--

R ² c (Fixed + Random effects): 0.4515
--

Random effects contribution: 0.4311
--

Visualizing Wealth Results for Video Stimuli

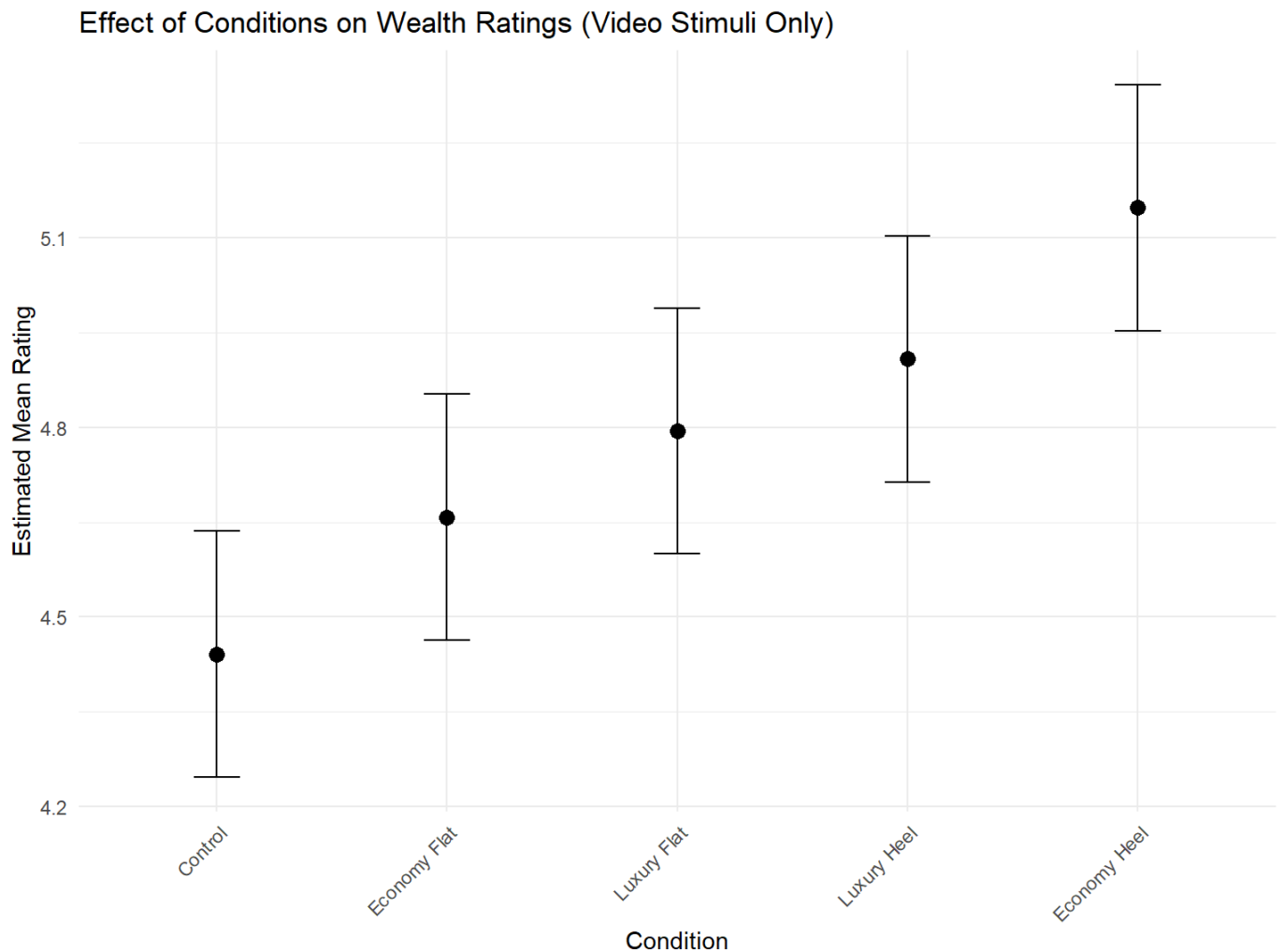


Figure B3.5. Effect of footwear conditions on wealth perception ratings for video stimuli, including control condition. Error bars represent 95% confidence intervals.

Pairwise Comparisons for Condition in Wealth Model (Video Stimuli)

```
## contrast          estimate    SE  df z.ratio p.value
## Control - Economy Flat    -0.217 0.0471 Inf  -4.600 <.0001
## Control - Economy Heel    -0.707 0.0472 Inf -14.984 <.0001
## Control - Luxury Flat     -0.353 0.0471 Inf  -7.495 <.0001
## Control - Luxury Heel     -0.467 0.0471 Inf  -9.906 <.0001
## Economy Flat - Economy Heel -0.490 0.0472 Inf -10.391 <.0001
## Economy Flat - Luxury Flat -0.137 0.0471 Inf  -2.897 0.0308
## Economy Flat - Luxury Heel -0.250 0.0471 Inf  -5.311 <.0001
## Economy Heel - Luxury Flat  0.353 0.0472 Inf   7.492 <.0001
## Economy Heel - Luxury Heel  0.240 0.0472 Inf   5.079 <.0001
## Luxury Flat - Luxury Heel  -0.114 0.0472 Inf  -2.413 0.1118
##
## Degrees-of-freedom method: asymptotic
## P value adjustment: tukey method for comparing a family of 5 estimates
```


Comparison Between Dimensions for Video Stimuli

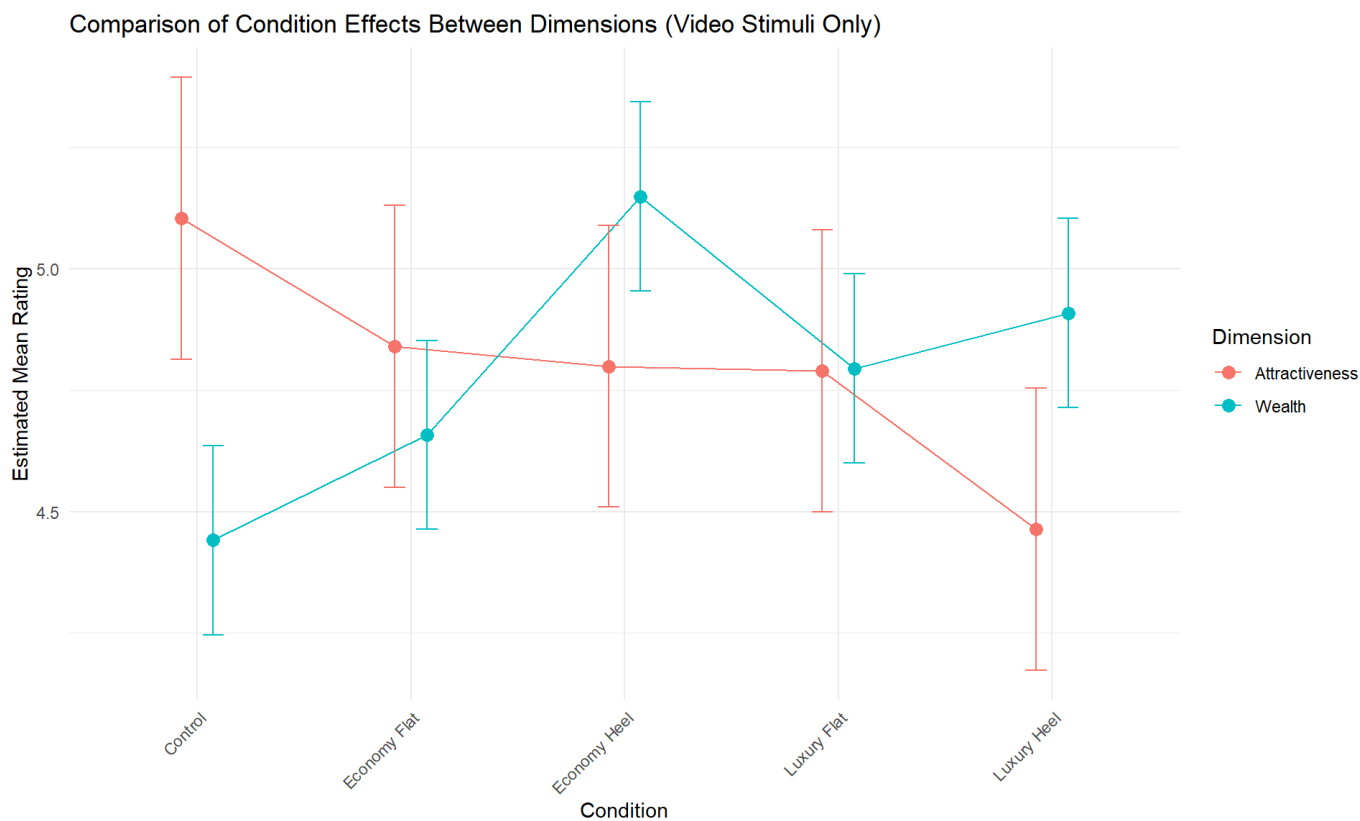
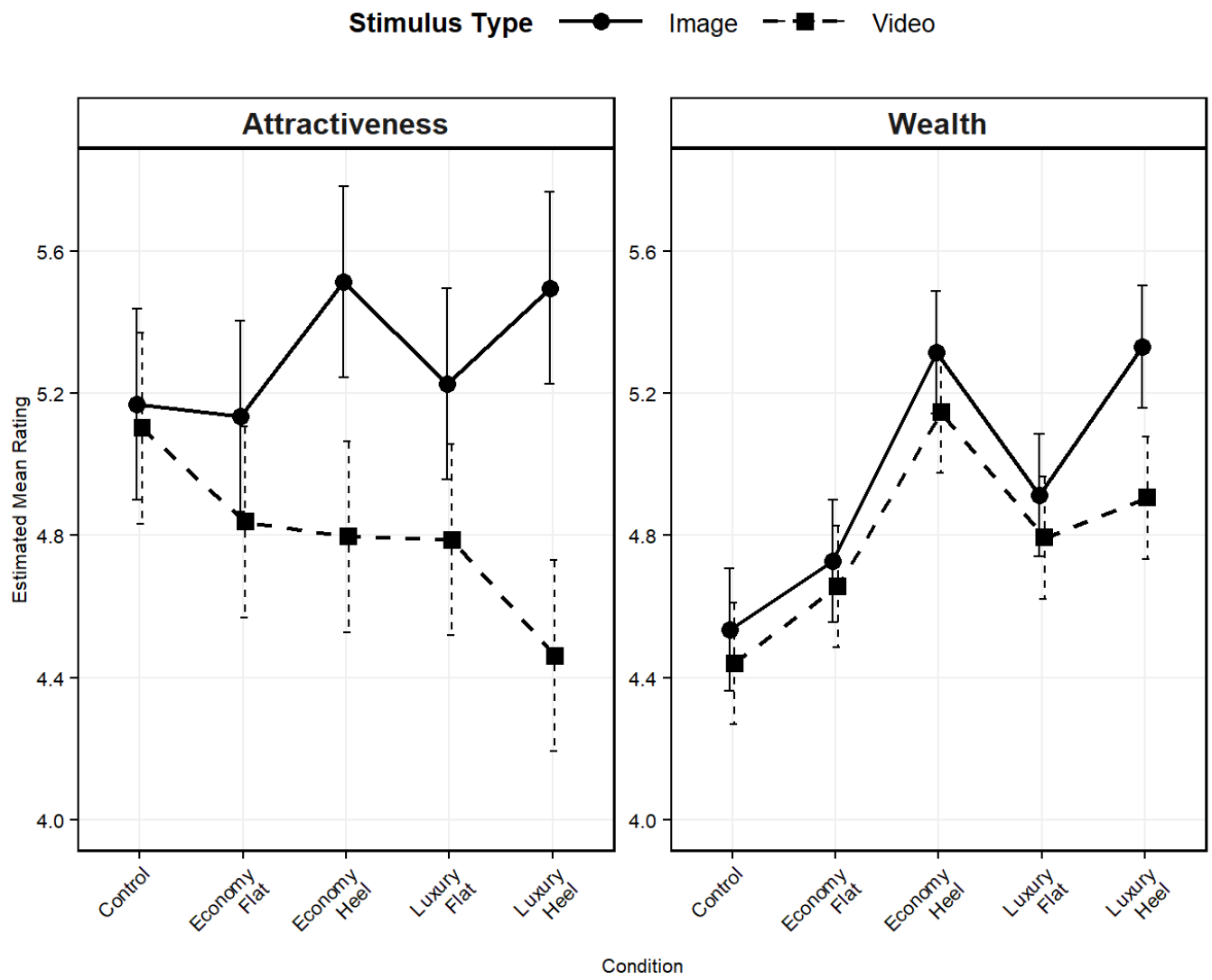


Figure B3.6. Comparison of footwear condition effects between attractiveness and wealth dimensions for video stimuli. Error bars represent 95% confidence intervals.

Creating comparison plot for Dimention x StimuliType



Error bars represent 95% confidence intervals

Figure B3.7. Comparison of footwear condition effects across stimulus types (image vs. video) and dimensions (attractiveness vs. wealth). Error bars represent 95% confidence intervals.

PART IV: Supplement B4: 2×2 Factorial Analysis (Excluding Control Condition)

13. Data Preparation for 2×2 Factorial Design

14. Attractiveness Models - 2×2 Factorial Design

14.1 Attractiveness Model for Image Stimuli (2×2)

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Trait_Exaggeration * Cost + (1 | block/Model) + (1 |  
## code)  
## Data: body_df_attract_factorial_image  
##  
## REML criterion at convergence: 18152.7  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -5.1205 -0.5987 -0.0033  0.5779  4.2893  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept)  2.0089    1.4173  
## Model:block (Intercept)  0.6718    0.8196  
## block        (Intercept)  0.0000    0.0000  
## Residual                    1.3251    1.1511  
## Number of obs: 5450, groups:  code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)      5.34189    0.14117  115.86546  37.840   <2e-16 ***  
## Trait_Exaggeration1 -0.16251    0.01559  5124.96037 -10.421   <2e-16 ***  
## Cost1             -0.01816    0.01559  5124.98257  -1.164    0.2443  
## Trait_Exaggeration1:Cost1 -0.02705    0.01559  5124.97924  -1.734    0.0829 .  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) Trt_E1 Cost1  
## Trt_Exggrt1  0.000  
## Cost1        0.000  0.002  
## Trt_Exg1:C1  0.000  0.000  0.000  
## optimizer (nloptwrap) convergence code: 0 (OK)  
## boundary (singular) fit: see help('isSingular')
```

ANOVA for 2×2 Factorial Model (Attractiveness - Image)

```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF DenDF  F value Pr(>F)
## Trait_Exaggeration    143.904  143.904      1  5125 108.6011 <2e-16 ***
## Cost                   1.797    1.797      1  5125   1.3560 0.2443
## Trait_Exaggeration:Cost    3.986    3.986      1  5125   3.0083 0.0829 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means (Attractiveness - Image)

```
## [1] "Main Effect of Trait Exaggeration:"
```

```
## Trait_Exaggeration emmean    SE  df asymp.LCL asymp.UCL
## Flat              5.18 0.142 Inf      4.90      5.46
## Heel              5.50 0.142 Inf      5.23      5.78
##
## Results are averaged over the levels of: Cost
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Main Effect of Cost:"
```

```
## Cost    emmean    SE  df asymp.LCL asymp.UCL
## Economy 5.32 0.142 Inf      5.05      5.60
## Luxury  5.36 0.142 Inf      5.08      5.64
##
## Results are averaged over the levels of: Trait_Exaggeration
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Interaction: Trait Exaggeration x Cost:"
```

```
## Cost = Economy:
## Trait_Exaggeration emmean    SE  df asymp.LCL asymp.UCL
## Flat              5.13 0.144 Inf      4.85      5.42
## Heel              5.51 0.144 Inf      5.23      5.79
##
## Cost = Luxury:
## Trait_Exaggeration emmean    SE  df asymp.LCL asymp.UCL
## Flat              5.22 0.144 Inf      4.94      5.51
## Heel              5.50 0.144 Inf      5.21      5.78
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Pairwise Comparisons (Attractiveness - Image)

```
## Cost = Economy:
## contrast      estimate      SE  df z.ratio p.value
## Flat - Heel   -0.379 0.0441 Inf  -8.595 <.0001
##
## Cost = Luxury:
## contrast      estimate      SE  df z.ratio p.value
## Flat - Heel   -0.271 0.0441 Inf  -6.142 <.0001
##
## Degrees-of-freedom method: asymptotic
```

```
## Trait_Exaggeration = Flat:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury -0.0904 0.0441 Inf  -2.050 0.0403
##
## Trait_Exaggeration = Heel:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury 0.0178 0.0441 Inf   0.403 0.6870
##
## Degrees-of-freedom method: asymptotic
```

R-squared for Attractiveness Model (Image - 2x2)

```
##              R2m      R2c
## [1,] 0.006810238 0.6714619
```

```
## Attractiveness Model (Image) R-squared:
```

```
## R2m (Fixed effects only): 0.0068
```

```
## R2c (Fixed + Random effects): 0.6715
```

```
## Random effects contribution: 0.6647
```

14.2 Attractiveness Model for Video Stimuli (2×2)

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Trait_Exaggeration * Cost + (1 | block/Model) + (1 |  
## code)  
## Data: body_df_attract_factorial_video  
##  
## REML criterion at convergence: 19364.4  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -3.6732 -0.6332 -0.0364  0.6063  3.8647  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept) 1.7309    1.3157  
## Model:block (Intercept) 0.7686    0.8767  
## block        (Intercept) 0.0000    0.0000  
## Residual                    1.6837    1.2976  
## Number of obs: 5454, groups: code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##                                Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)                   4.72322    0.14418   96.83055  32.759 < 2e-16 ***  
## Trait_Exaggeration1            0.09129    0.01757  5129.22221   5.195 2.12e-07 ***  
## Cost1                         0.09629    0.01757  5129.18611   5.480 4.45e-08 ***  
## Trait_Exaggeration1:Cost1     -0.07103    0.01757  5129.16713  -4.042 5.37e-05 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) Trt_E1 Cost1  
## Trt_Exggrt1  0.000  
## Cost1        0.000  0.000  
## Trt_Exg1:C1  0.000 -0.001 -0.001  
## optimizer (nloptwrap) convergence code: 0 (OK)  
## boundary (singular) fit: see help('isSingular')
```

ANOVA for 2×2 Factorial Model (Attractiveness - Video)

```
## Type III Analysis of Variance Table with Satterthwaite's method  
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)  
## Trait_Exaggeration  45.445  45.445     1 5129.2  26.992 2.122e-07 ***  
## Cost                50.564  50.564     1 5129.2  30.032 4.453e-08 ***  
## Trait_Exaggeration:Cost 27.514  27.514     1 5129.2  16.342 5.367e-05 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means (Attractiveness - Video)

```
## [1] "Main Effect of Trait Exaggeration:"
```

```
## Trait_Exaggeration emmean    SE  df asymp.LCL asymp.UCL
## Flat                4.81 0.145 Inf      4.53      5.10
## Heel                4.63 0.145 Inf      4.35      4.92
##
## Results are averaged over the levels of: Cost
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Main Effect of Cost:"
```

```
## Cost      emmean    SE  df asymp.LCL asymp.UCL
## Economy   4.82 0.145 Inf      4.53      5.10
## Luxury    4.63 0.145 Inf      4.34      4.91
##
## Results are averaged over the levels of: Trait_Exaggeration
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Interaction: Trait Exaggeration × Cost:"
```

```
## Cost = Economy:
## Trait_Exaggeration emmean    SE  df asymp.LCL asymp.UCL
## Flat                4.84 0.147 Inf      4.55      5.13
## Heel                4.80 0.147 Inf      4.51      5.09
##
## Cost = Luxury:
## Trait_Exaggeration emmean    SE  df asymp.LCL asymp.UCL
## Flat                4.79 0.147 Inf      4.50      5.08
## Heel                4.46 0.147 Inf      4.18      4.75
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Pairwise Comparisons (Attractiveness - Video)

```
## Cost = Economy:
## contrast      estimate    SE  df z.ratio p.value
## Flat - Heel   0.0405 0.0497 Inf   0.816  0.4146
##
## Cost = Luxury:
## contrast      estimate    SE  df z.ratio p.value
## Flat - Heel   0.3247 0.0497 Inf   6.530  <.0001
##
## Degrees-of-freedom method: asymptotic
```

```
## Trait_Exaggeration = Flat:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury  0.0505 0.0497 Inf   1.017  0.3091
##
## Trait_Exaggeration = Heel:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury  0.3347 0.0497 Inf   6.731  <.0001
##
## Degrees-of-freedom method: asymptotic
```

R-squared for Attractiveness Model (Video - 2×2)

```
##           R2m      R2c
## [1,] 0.00538388 0.5996845
```

```
## Attractiveness Model (Video) R-squared:
```

```
## R2m (Fixed effects only): 0.0054
```

```
## R2c (Fixed + Random effects): 0.5997
```

```
## Random effects contribution: 0.5943
```


15. Wealth Models - 2×2 Factorial Design

15.1 Wealth Model for Image Stimuli (2×2)

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: Rating ~ Trait_Exaggeration * Cost + (1 | block/Model) + (1 |
## code)
## Data: body_df_wealth_factorial_image
##
## REML criterion at convergence: 16675.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.4860 -0.5730 -0.0212  0.5351  4.3410
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## code        (Intercept)  0.9865     0.9932
## Model:block (Intercept)  0.1999     0.4471
## block        (Intercept)  0.0000     0.0000
## Residual                    1.0522     1.0258
## Number of obs: 5433, groups: code, 279; Model:block, 54; block, 11
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      5.07080    0.08631 152.09956  58.752 < 2e-16 ***
## Trait_Exaggeration1 -0.25202    0.01392 5108.06769 -18.106 < 2e-16 ***
## Cost1             -0.05004    0.01392 5108.09220  -3.595 0.000327 ***
## Trait_Exaggeration1:Cost1 -0.04224    0.01392 5108.11232  -3.035 0.002419 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) Trt_E1 Cost1
## Trt_Exggrt1  0.000
## Cost1        0.000  0.002
## Trt_Exg1:C1  0.000 -0.001 -0.002
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

ANOVA for 2×2 Factorial Model (Wealth - Image)

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF  F value    Pr(>F)
## Trait_Exaggeration    344.94   344.94     1 5108.1 327.8235 < 2.2e-16 ***
## Cost                  13.60    13.60     1 5108.1 12.9243 0.0003274 ***
## Trait_Exaggeration:Cost   9.69     9.69     1 5108.1  9.2104 0.0024186 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means (Wealth - Image)

```
## [1] "Main Effect of Trait Exaggeration:"
```

```
## Trait_Exaggeration emmean      SE  df asymp.LCL asymp.UCL
## Flat                4.82 0.0874 Inf      4.65      4.99
## Heel                5.32 0.0874 Inf      5.15      5.49
##
## Results are averaged over the levels of: Cost
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Main Effect of Cost:"
```

```
## Cost      emmean      SE  df asymp.LCL asymp.UCL
## Economy   5.02 0.0874 Inf      4.85      5.19
## Luxury    5.12 0.0874 Inf      4.95      5.29
##
## Results are averaged over the levels of: Trait_Exaggeration
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Interaction: Trait Exaggeration × Cost:"
```

```
## Cost = Economy:
## Trait_Exaggeration emmean      SE  df asymp.LCL asymp.UCL
## Flat                4.73 0.0896 Inf      4.55      4.90
## Heel                5.32 0.0896 Inf      5.14      5.49
##
## Cost = Luxury:
## Trait_Exaggeration emmean      SE  df asymp.LCL asymp.UCL
## Flat                4.91 0.0896 Inf      4.74      5.09
## Heel                5.33 0.0896 Inf      5.15      5.51
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Pairwise Comparisons (Wealth - Image)

```
## Cost = Economy:
## contrast      estimate      SE  df z.ratio p.value
## Flat - Heel   -0.589 0.0394 Inf -14.953 <.0001
##
## Cost = Luxury:
## contrast      estimate      SE  df z.ratio p.value
## Flat - Heel   -0.420 0.0394 Inf -10.654 <.0001
##
## Degrees-of-freedom method: asymptotic
```

```
## Trait_Exaggeration = Flat:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury -0.1846 0.0393 Inf  -4.692 <.0001
##
## Trait_Exaggeration = Heel:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury -0.0156 0.0394 Inf  -0.396 0.6923
##
## Degrees-of-freedom method: asymptotic
```

R-squared for Wealth Model (Image - 2×2)

```
##           R2m      R2c
## [1,] 0.02939144 0.5437916
```

```
## Wealth Model (Image) R-squared:
```

```
## R2m (Fixed effects only): 0.0294
```

```
## R2c (Fixed + Random effects): 0.5438
```

```
## Random effects contribution: 0.5144
```

15.2 Wealth Model for Video Stimuli (2×2)

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: Rating ~ Trait_Exaggeration * Cost + (1 | block/Model) + (1 |  
## code)  
## Data: body_df_wealth_factorial_video  
##  
## REML criterion at convergence: 18551.4  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -4.3298 -0.5933 -0.0255  0.5700  5.0498  
##  
## Random effects:  
## Groups      Name                Variance Std.Dev.  
## code        (Intercept) 0.9714    0.9856  
## Model:block (Intercept) 0.3069    0.5540  
## block       (Intercept) 0.0000    0.0000  
## Residual                    1.4999    1.2247  
## Number of obs: 5446, groups: code, 279; Model:block, 54; block, 11  
##  
## Fixed effects:  
##                                Estimate Std. Error      df t value Pr(>|t|)  
## (Intercept)                   4.87793    0.09727  114.55841  50.147 < 2e-16 ***  
## Trait_Exaggeration1            -0.15130    0.01660  5121.41542  -9.116 < 2e-16 ***  
## Cost1                          0.02555    0.01660  5121.36604   1.540  0.124  
## Trait_Exaggeration1:Cost1      -0.09381    0.01660  5121.36349  -5.652 1.67e-08 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) Trt_E1 Cost1  
## Trt_Exggrt1  0.000  
## Cost1        0.000 -0.001  
## Trt_Exg1:C1  0.000  0.000 -0.001  
## optimizer (nloptwrap) convergence code: 0 (OK)  
## boundary (singular) fit: see help('isSingular')
```

ANOVA for 2×2 Factorial Model (Wealth - Video)

```
## Type III Analysis of Variance Table with Satterthwaite's method  
##                                Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)  
## Trait_Exaggeration            124.633  124.633     1  5121.4 83.0971 < 2.2e-16 ***  
## Cost                          3.555    3.555     1  5121.4  2.3705  0.1237  
## Trait_Exaggeration:Cost       47.914   47.914     1  5121.4 31.9460 1.671e-08 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Estimated Marginal Means (Wealth - Video)

```
## [1] "Main Effect of Trait Exaggeration:"
```

```
## Trait_Exaggeration emmean      SE  df asymp.LCL asymp.UCL
## Flat                4.73 0.0987 Inf      4.53      4.92
## Heel                5.03 0.0987 Inf      4.84      5.22
##
## Results are averaged over the levels of: Cost
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Main Effect of Cost:"
```

```
## Cost      emmean      SE  df asymp.LCL asymp.UCL
## Economy   4.90 0.0987 Inf      4.71      5.10
## Luxury    4.85 0.0987 Inf      4.66      5.05
##
## Results are averaged over the levels of: Trait_Exaggeration
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
## [1] "Interaction: Trait Exaggeration × Cost:"
```

```
## Cost = Economy:
## Trait_Exaggeration emmean      SE  df asymp.LCL asymp.UCL
## Flat                4.66 0.101 Inf      4.46      4.86
## Heel                5.15 0.101 Inf      4.95      5.35
##
## Cost = Luxury:
## Trait_Exaggeration emmean      SE  df asymp.LCL asymp.UCL
## Flat                4.79 0.101 Inf      4.60      4.99
## Heel                4.91 0.101 Inf      4.71      5.11
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

Pairwise Comparisons (Wealth - Video)

```
## Cost = Economy:
## contrast      estimate      SE  df z.ratio p.value
## Flat - Heel   -0.490 0.0469 Inf -10.443 <.0001
##
## Cost = Luxury:
## contrast      estimate      SE  df z.ratio p.value
## Flat - Heel   -0.115 0.0469 Inf  -2.449 0.0143
##
## Degrees-of-freedom method: asymptotic
```

```
## Trait_Exaggeration = Flat:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury  -0.137 0.0469 Inf  -2.910 0.0036
##
## Trait_Exaggeration = Heel:
## contrast      estimate      SE  df z.ratio p.value
## Economy - Luxury   0.239 0.0470 Inf   5.082 <.0001
##
## Degrees-of-freedom method: asymptotic
```

R-squared for Wealth Model (Video - 2×2)

```
##           R2m      R2c
## [1,] 0.01150715 0.4663404
```

```
## Wealth Model (Video) R-squared:
```

```
## R2m (Fixed effects only): 0.0115
```

```
## R2c (Fixed + Random effects): 0.4663
```

```
## Random effects contribution: 0.4548
```

16. Visualization of 2×2 Factorial Effects

16.1 Interaction Plots for All Models

2×2 Factorial Effects: Trait Exaggeration × Cost

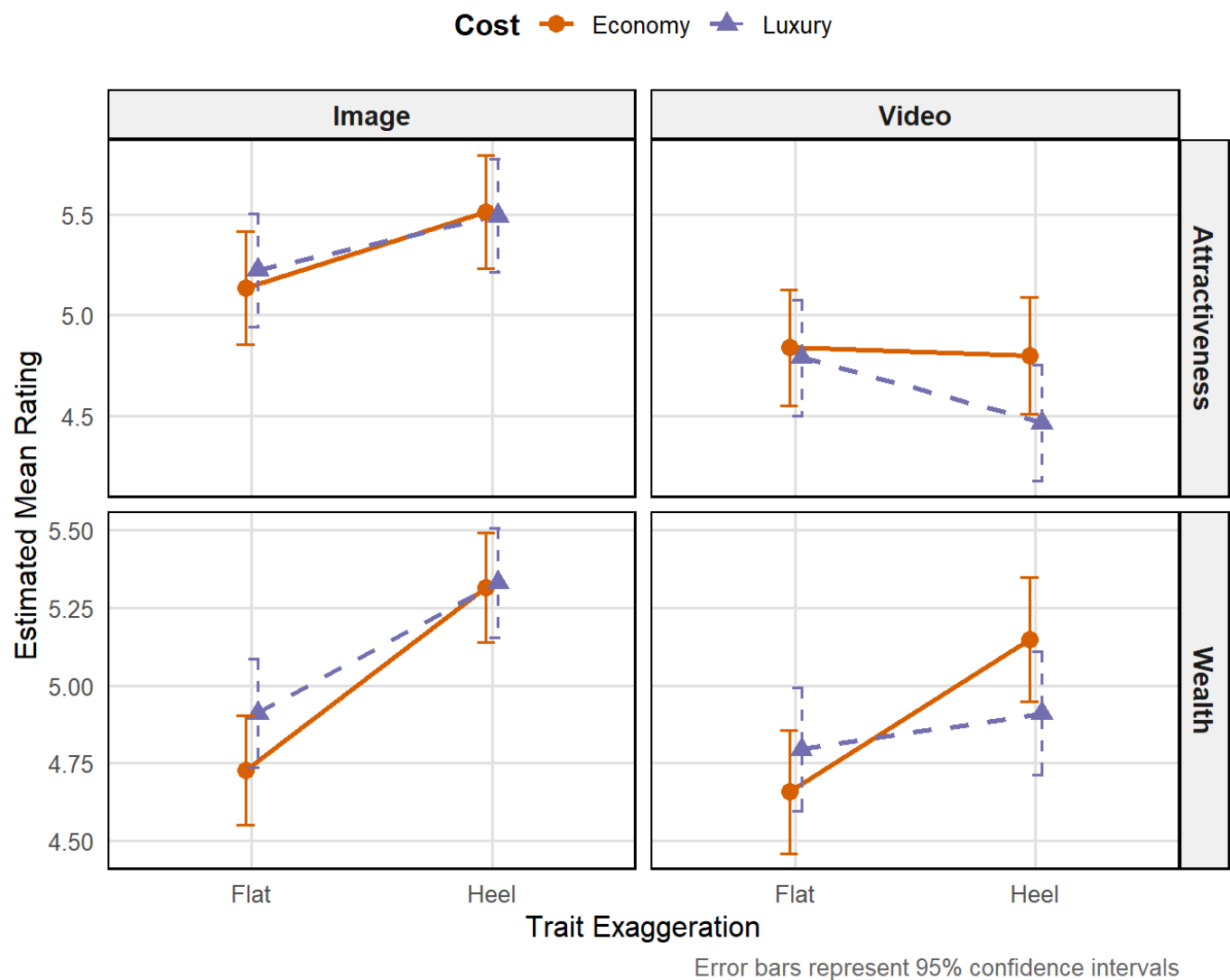


Figure B4.1. Interaction between trait exaggeration (heel vs. flat) and cost (economy vs. luxury) across stimulus types and rating dimensions. Error bars represent 95% confidence intervals.

16.2 Main Effects Plots

Main Effects of Trait Exaggeration and Cost

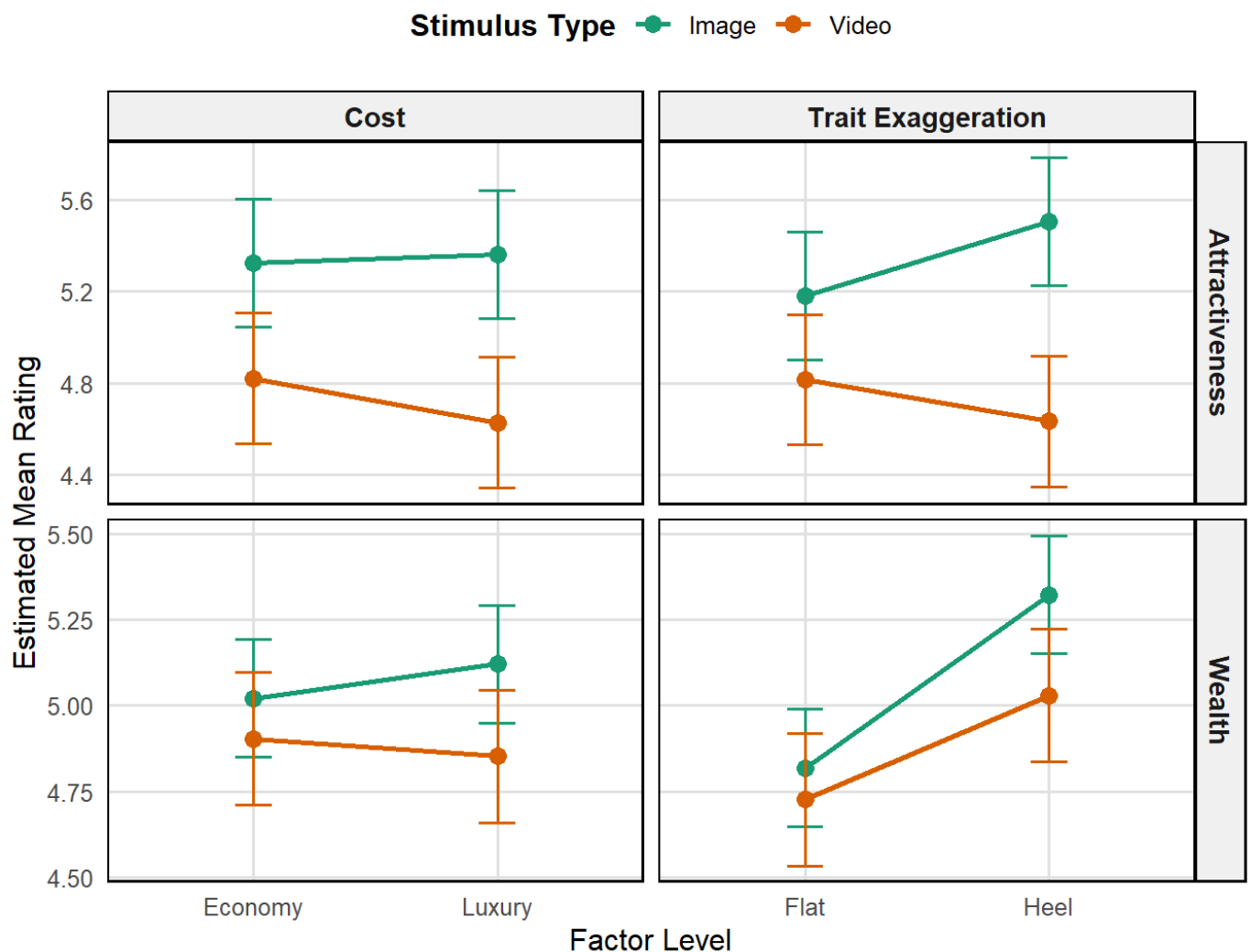


Figure B4.2. Main effects of cost and trait exaggeration across stimulus types for attractiveness and wealth ratings. Error bars represent 95% confidence intervals.

17. Effect Size Analysis for 2×2 Factorial Models

17.1. Function to Calculate Cohen's d from emmeans

17.2 Calculate Effect Sizes for All Models

```
##  
## === ATTRACTIVENESS - IMAGE STIMULI ===
```

```
##  
## Main Effect: Trait Exaggeration
```

```
##      contrast  estimate  cohens_d effect_size_label  p.value  
## 1 Flat - Heel -0.3250175 -0.2823501      Small 1.984627e-25
```



```
##
## Main Effect: Cost
```

```
##          contrast    estimate    cohens_d effect_size_label    p.value
## 1 Economy - Luxury -0.03631908 -0.03155122      Very Small 0.2442293
```

```
##
## Simple Effects: Trait Exaggeration within Cost levels
```

```
##          contrast    estimate    cohens_d effect_size_label    p.value
## 1 Flat - Heel -0.3791129 -0.3293441      Small 8.314086e-18
## 2 Flat - Heel -0.2709220 -0.2353562      Small 8.128295e-10
```

```
##
##
## === ATTRACTIVENESS - VIDEO STIMULI ===
```

```
##
## Main Effect: Trait Exaggeration
```

```
##          contrast    estimate    cohens_d effect_size_label    p.value
## 1 Flat - Heel 0.1825885 0.1407169      Very Small 2.042923e-07
```

```
##
## Main Effect: Cost
```

```
##          contrast    estimate    cohens_d effect_size_label    p.value
## 1 Economy - Luxury 0.1925889 0.148424      Very Small 4.249587e-08
```

```
##
## Simple Effects: Trait Exaggeration within Cost levels
```

```
##          contrast    estimate    cohens_d effect_size_label    p.value
## 1 Flat - Heel 0.04052668 0.03123301      Very Small 4.146398e-01
## 2 Flat - Heel 0.32465027 0.25020076      Small 6.598126e-11
```

```
##
##
## === WEALTH - IMAGE STIMULI ===
```

```
##
## Main Effect: Trait Exaggeration
```

```
##          contrast    estimate    cohens_d effect_size_label    p.value
## 1 Flat - Heel -0.504031 -0.4913652      Medium 2.863108e-73
```

```
##
## Main Effect: Cost
```

```
##          contrast  estimate  cohens_d effect_size_label  p.value
## 1 Economy - Luxury -0.1000797 -0.09756482      Very Small 0.0003243445
```

```
##
## Simple Effects: Trait Exaggeration within Cost levels
```

```
##          contrast  estimate  cohens_d effect_size_label  p.value
## 1 Flat - Heel -0.5885178 -0.5737290      Medium 1.492362e-50
## 2 Flat - Heel -0.4195441 -0.4090014      Medium 1.677902e-26
```

```
##
##
## === WEALTH - VIDEO STIMULI ===
```

```
##
## Main Effect: Trait Exaggeration
```

```
##          contrast  estimate  cohens_d effect_size_label  p.value
## 1 Flat - Heel -0.3026059 -0.247089      Small 7.812207e-20
```

```
##
## Main Effect: Cost
```

```
##          contrast  estimate  cohens_d effect_size_label  p.value
## 1 Economy - Luxury 0.05110851 0.04173199      Very Small 0.1236474
```

```
##
## Simple Effects: Trait Exaggeration within Cost levels
```

```
##          contrast  estimate  cohens_d effect_size_label  p.value
## 1 Flat - Heel -0.4902260 -0.40028769      Medium 1.580942e-25
## 2 Flat - Heel -0.1149859 -0.09389024      Very Small 1.431416e-02
```

17.3 Partial Eta-squared from ANOVA Tables

```
##
## === Attractiveness - Image ===
##              Sum Sq Mean Sq NumDF DenDF  F value    Pr(>F)
## Trait_Exaggeration    143.904  143.904      1  5125 108.6011 0.00000
## Cost                   1.797    1.797      1  5125   1.3560 0.24428
## Trait_Exaggeration:Cost    3.986    3.986      1  5125   3.0083 0.08290
##              partial_eta_sq effect_size_label
## Trait_Exaggeration         0.0207122              2
## Cost                       0.0002640              1
## Trait_Exaggeration:Cost     0.0005855              1
```

```
##
## === Attractiveness - Video ===
##              Sum Sq Mean Sq NumDF  DenDF F value      Pr(>F)
## Trait_Exaggeration    45.445  45.445      1 5129.2  26.992 2.1200e-07
## Cost                  50.564  50.564      1 5129.2  30.032 4.5000e-08
## Trait_Exaggeration:Cost 27.514  27.514      1 5129.2  16.342 5.3666e-05
##              partial_eta_sq effect_size_label
## Trait_Exaggeration         0.0052211              1
## Cost                       0.0058057              1
## Trait_Exaggeration:Cost     0.0031675              1
```

```
##
## === Wealth - Image ===
##              Sum Sq Mean Sq NumDF  DenDF  F value      Pr(>F)
## Trait_Exaggeration    344.94  344.94      1 5108.1 327.8235 0.00000000
## Cost                  13.60   13.60      1 5108.1  12.9243 0.00032741
## Trait_Exaggeration:Cost   9.69    9.69      1 5108.1   9.2104 0.00241860
##              partial_eta_sq effect_size_label
## Trait_Exaggeration         0.060127              3
## Cost                       0.002516              1
## Trait_Exaggeration:Cost     0.001794              1
```

```
##
## === Wealth - Video ===
##              Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## Trait_Exaggeration    124.633  124.633      1 5121.4 83.0971 0.00000
## Cost                   3.555    3.555      1 5121.4   2.3705 0.12371
## Trait_Exaggeration:Cost 47.914  47.914      1 5121.4 31.9460 0.00000
##              partial_eta_sq effect_size_label
## Trait_Exaggeration         0.0159004              2
## Cost                       0.0004607              1
## Trait_Exaggeration:Cost     0.0061732              1
```

17.4 Comprehensive Summary Table of Effect Sizes

##	Model	Effect	Raw_Difference	Cohens_d
## 1	Attractiveness - Image Trait Exaggeration (Main)		0.325	0.28
## 2	Attractiveness - Image Cost (Main)		0.036	0.03
## 3	Attractiveness - Image Trait × Cost (Interaction)		NA	NA
## 4	Attractiveness - Video Trait Exaggeration (Main)		-0.183	-0.14
## 5	Attractiveness - Video Cost (Main)		-0.193	-0.15
## 6	Attractiveness - Video Trait × Cost (Interaction)		NA	NA
## 7	Wealth - Image Trait Exaggeration (Main)		0.504	0.49
## 8	Wealth - Image Cost (Main)		0.100	0.10
## 9	Wealth - Image Trait × Cost (Interaction)		NA	NA
## 10	Wealth - Video Trait Exaggeration (Main)		0.302	0.25
## 11	Wealth - Video Cost (Main)		-0.051	-0.04
## 12	Wealth - Video Trait × Cost (Interaction)		NA	NA
##	Partial_Eta_Sq			
## 1	0.0207			
## 2	0.0003			
## 3	0.0006			
## 4	0.0052			
## 5	0.0058			
## 6	0.0032			
## 7	0.0601			
## 8	0.0025			
## 9	0.0018			
## 10	0.0159			
## 11	0.0005			
## 12	0.0062			

17.5 Simple Effects Summary for Key Comparisons

##	Model	Comparison	Raw_Difference_Heel_Higher
## 1	Attractiveness - Image: Economy	Heel vs Flat	0.379
## 2	Attractiveness - Image: Luxury	Heel vs Flat	0.271
## 3	Attractiveness - Video: Economy	Heel vs Flat	-0.041
## 4	Attractiveness - Video: Luxury	Heel vs Flat	-0.325
## 5	Wealth - Image: Economy	Heel vs Flat	0.589
## 6	Wealth - Image: Luxury	Heel vs Flat	0.420
## 7	Wealth - Video: Economy	Heel vs Flat	0.490
## 8	Wealth - Video: Luxury	Heel vs Flat	0.115

##	Cohens_d_abs	Effect_Size_Label	p_value
## 1	0.33	Small	8.31e-18
## 2	0.24	Small	8.13e-10
## 3	0.03	Negligible	4.15e-01
## 4	0.25	Small	6.60e-11
## 5	0.57	Medium	1.49e-50
## 6	0.41	Small	1.68e-26
## 7	0.40	Small	1.58e-25
## 8	0.09	Negligible	1.43e-02

##	Interpretation
## 1	Heels INCREASE attractiveness noticeably in images (Economy)
## 2	Heels INCREASE attractiveness noticeably in images (Luxury)
## 3	No meaningful difference in videos (Economy)
## 4	Heels INCREASE attractiveness slightly in videos (Luxury)
## 5	Heels SUBSTANTIALY INCREASE wealth perception in images (Economy)
## 6	Heels INCREASE wealth perception in images (Luxury)
## 7	Heels INCREASE wealth perception in videos (Economy)
## 8	Minimal difference in videos (Luxury)

17.6 Visualization: Effect Sizes Across Models

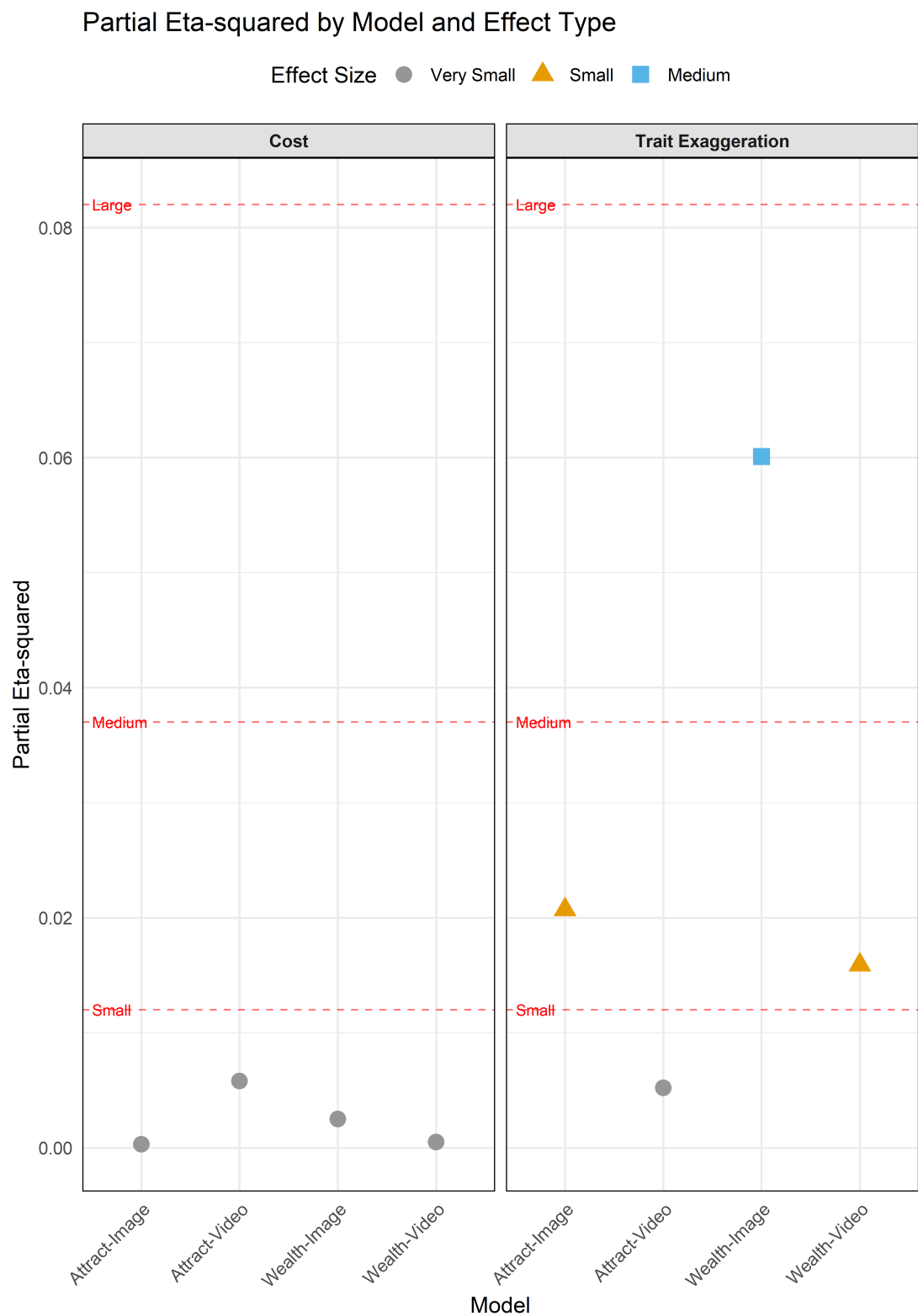


Figure B4.3. Partial Eta-squared effect sizes for main effects across all 2×2 factorial models. Dashed reference lines indicate small (0.012), medium (0.037), and large (0.082) effect size thresholds. For Effect Size Interpretation Guidelines see Supplement Section A3.

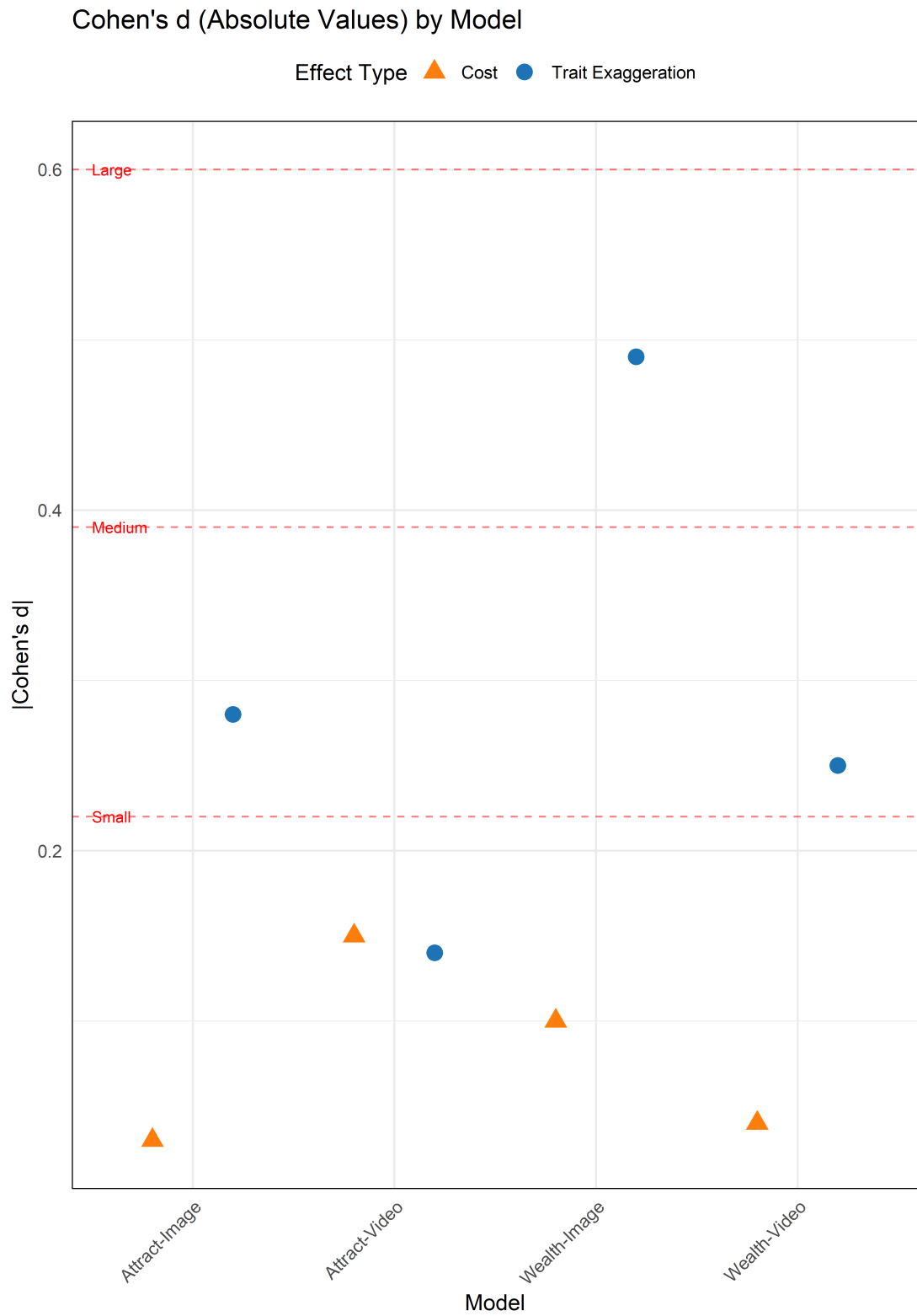


Figure B4.4. Cohen's d effect sizes (absolute values) for cost and trait exaggeration main effects across all models. Dashed reference lines indicate small (0.22), medium (0.39), and large (0.60) effect size thresholds. For Effect Size Interpretation Guidelines see Supplement Section A3.

Effect Sizes by Model and Metric

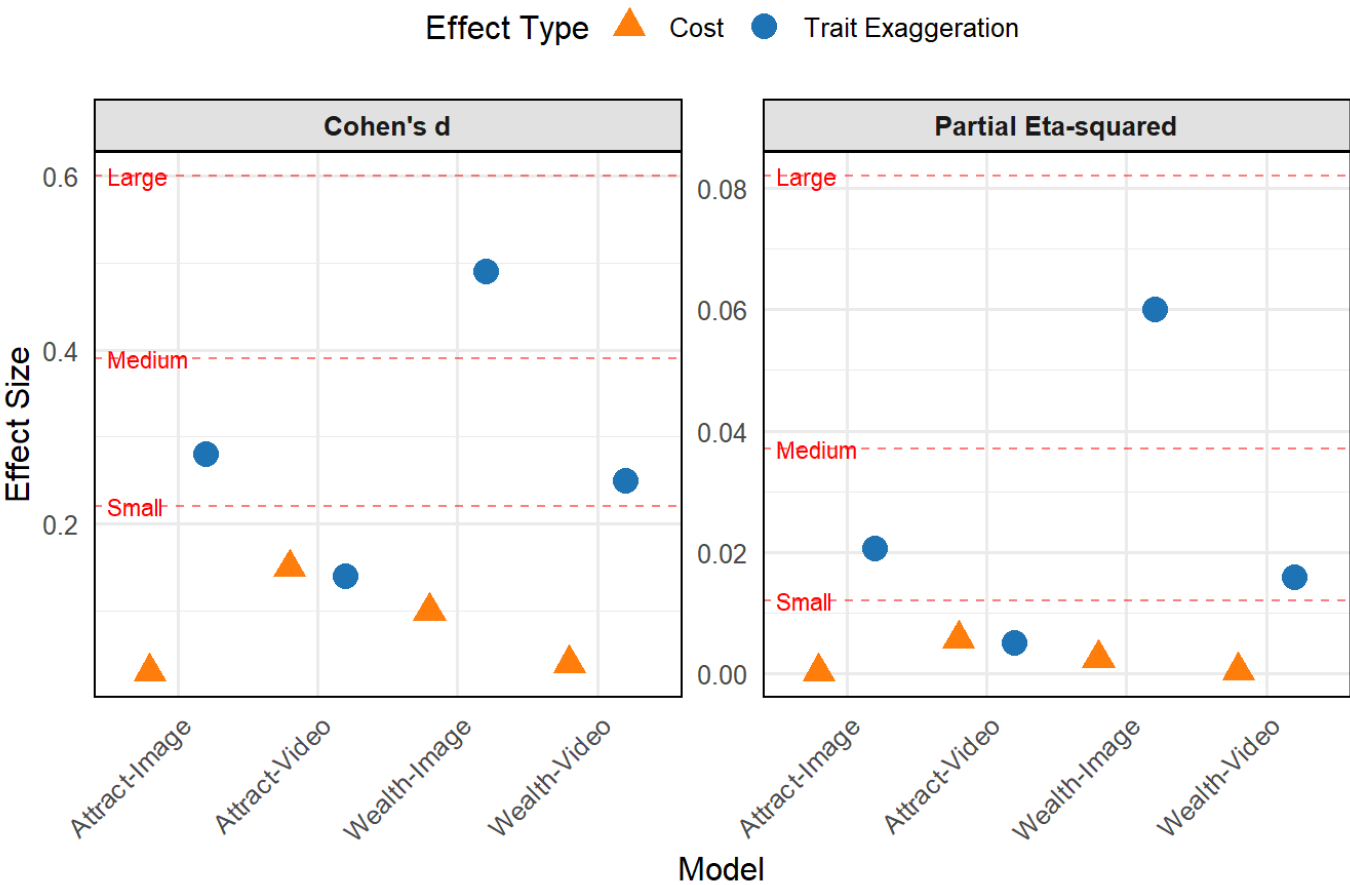


Figure B4.5. Effect sizes for main effects across all 2×2 factorial models, shown as Cohen’s d (left panel) and partial Eta-squared (right panel). Dashed reference lines indicate small, medium, and large effect size thresholds for each metric (d: 0.22, 0.39, 0.60; Eta-squared: 0.012, 0.037, 0.082). For Effect Size Interpretation Guidelines see Supplement Section A3.

18. Variance Components Analysis

```
##
## =====

## VARIANCE COMPONENTS ANALYSIS

## =====

## ATTRACTIVENESS MODEL VARIANCE COMPONENTS:

##      Component  Variance Percentage
## 5      Residual  1.7539217      41.90
## 1         code  1.7190197      41.06
## 2  Model:block  0.6135936      14.66
## 4 Fixed Effects 0.0996007       2.38
## 3         block 0.0000000       0.00
```


Percentage of variation explained by:

- Individual differences between raters: 41.06 %

- Stimuli structure (block/Model nested): 14.66 %

- Fixed effects (experimental conditions): 2.38 %

- Unexplained variation (residual): 41.9 %

WEALTH MODEL VARIANCE COMPONENTS:

##	Component	Variance	Percentage
## 5	Residual	1.40660054	55.45
## 1	code	0.84979078	33.50
## 2	Model:block	0.19431623	7.66
## 4	Fixed Effects	0.08589488	3.39
## 3	block	0.00000000	0.00

Percentage of variation explained by:

- Individual differences between raters: 33.5 %

- Stimuli structure (block/Model nested): 7.66 %

- Fixed effects (experimental conditions): 3.39 %

- Unexplained variation (residual): 55.45 %

ATTRACTIVENESS MODEL - IMAGES ONLY:

##	Component	Variance	Percentage
## 1	code	1.98903577	49.37
## 5	Residual	1.33634482	33.17
## 2	Model:block	0.67703476	16.80
## 4	Fixed Effects	0.02673105	0.66
## 3	block	0.00000000	0.00

ATTRACTIVENESS MODEL - VIDEOS ONLY:

```
##          Component    Variance Percentage
## 5      Residual 1.71108107      40.37
## 1          code 1.70473957      40.22
## 2    Model:block 0.78115677      18.43
## 4 Fixed Effects 0.04141337       0.98
## 3          block 0.00000000       0.00
```

```
##
## WEALTH MODEL - IMAGES ONLY:
```

```
##          Component    Variance Percentage
## 5      Residual 1.077144e+00      46.60
## 1          code 9.402658e-01      40.68
## 2    Model:block 1.935133e-01       8.37
## 4 Fixed Effects 1.004498e-01       4.35
## 3          block 1.581079e-09       0.00
```

```
##
## WEALTH MODEL - VIDEOS ONLY:
```

```
##          Component    Variance Percentage
## 5      Residual 1.513379e+00      54.85
## 1          code 8.904654e-01      32.27
## 2    Model:block 2.990295e-01      10.84
## 4 Fixed Effects 5.622153e-02       2.04
## 3          block 2.930203e-08       0.00
```

```
##
## =====
```

```
## VARIANCE COMPONENTS SUMMARY ACROSS MODELS
```

```
## =====
```

```
##          Model Rater_Variance Stimuli_Variance Fixed_Effects Residual
## 1 Attract - Full      41.06      14.66      2.38    41.90
## 2 Attract - Image     49.37      16.80      0.66    33.17
## 3 Attract - Video     40.22      18.43      0.98    40.37
## 4 Wealth - Full       33.50       7.66      3.39    55.45
## 5 Wealth - Image     40.68       8.37      4.35    46.60
## 6 Wealth - Video     32.27     10.84      2.04    54.85
```

PART V: Supplement B5: Normality Assumption Checks for Mixed-Effects Models with Nested Structure

20. Function to calculate standardized residuals (Pearson residuals scaled by their standard deviation)

21. Function to Check normality of random intercepts for each grouping factor

22. Function to perform formal normality tests

23. applying the functions to the models from section III

```
## === ATTRACTIVENESS IMAGE MODEL NORMALITY CHECKS ===
```

```
## Note: Shapiro-Wilk performed on random sample of 5000 residuals  
## Random effects for block have zero variance (boundary fit) - skipping normality test
```

```
##  
## === WEALTH MODEL NORMALITY CHECKS ===
```

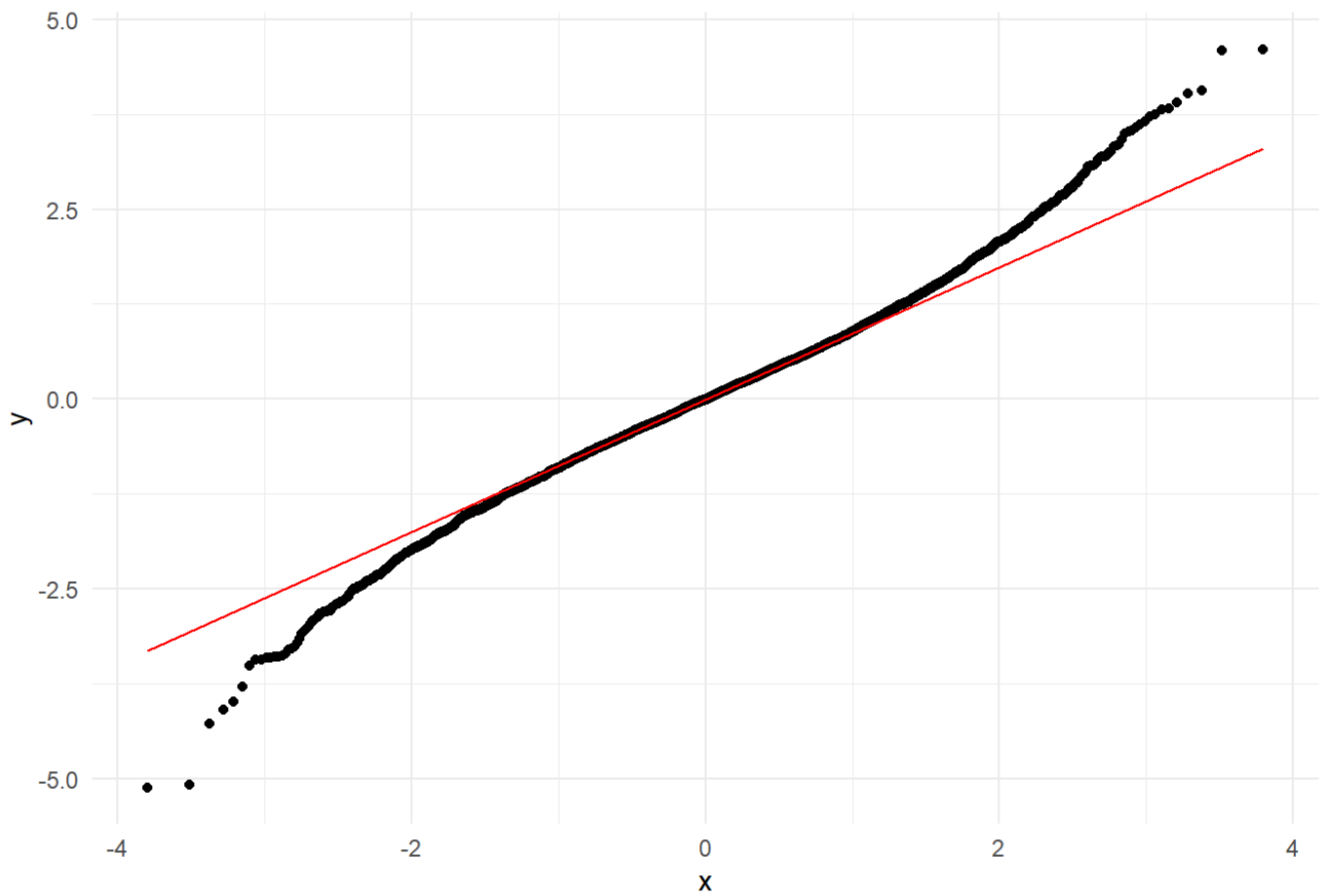
```
## Note: Shapiro-Wilk performed on random sample of 5000 residuals  
## Random effects for block have zero variance (boundary fit) - skipping normality test
```

24. Visualizing the results

```
## Model: Attractiveness, Image  
## Residuals Shapiro-Wilk: W = 0.9901 , p = 0  
## Residuals Anderson-Darling: A = 11.1158 , p = 0  
## Random Effects Tests:  
##   code : W = 0.9926 , p = 0.1828  
##   Model:block : W = 0.9782 , p = 0.4253  
##   block : Test not performed - Zero variance
```

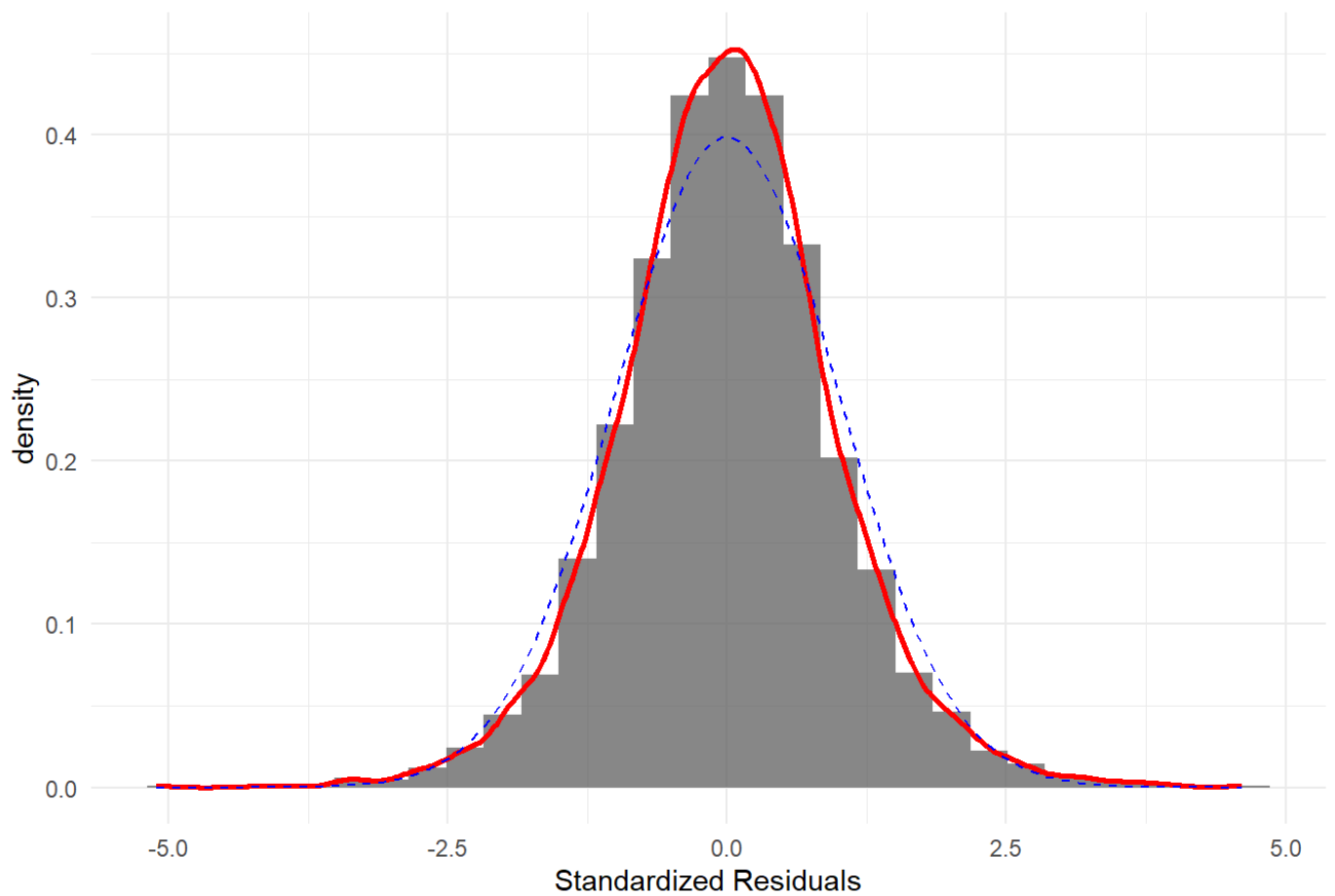
```
## Model: Wealth, Image  
## Residuals Shapiro-Wilk: W = 0.9867 , p = 0  
## Residuals Anderson-Darling: A = 22.2178 , p = 0  
## Random Effects Tests:  
##   code : W = 0.9833 , p = 0.0024  
##   Model:block : W = 0.9833 , p = 0.6496  
##   block : Test not performed - Zero variance
```

Q-Q Plot: Attractiveness, Image - Standardized Residuals



```
## Figure B5.1. Q-Q plot of standardized residuals for attractiveness ratings (image stimuli).  
## Points closely following the reference line (red),  
## indicate approximate normality of residuals.
```

Histogram: Attractiveness, Image - Standardized Residuals



```
## Figure B5.2. Histogram of standardized residuals for attractiveness ratings (image stimuli).  
## Red line shows kernel density estimate;  
## blue dashed line shows theoretical normal distribution ( $\mu=0$ ,  $\sigma=1$ ).
```

Residuals vs Fitted: Attractiveness, Image

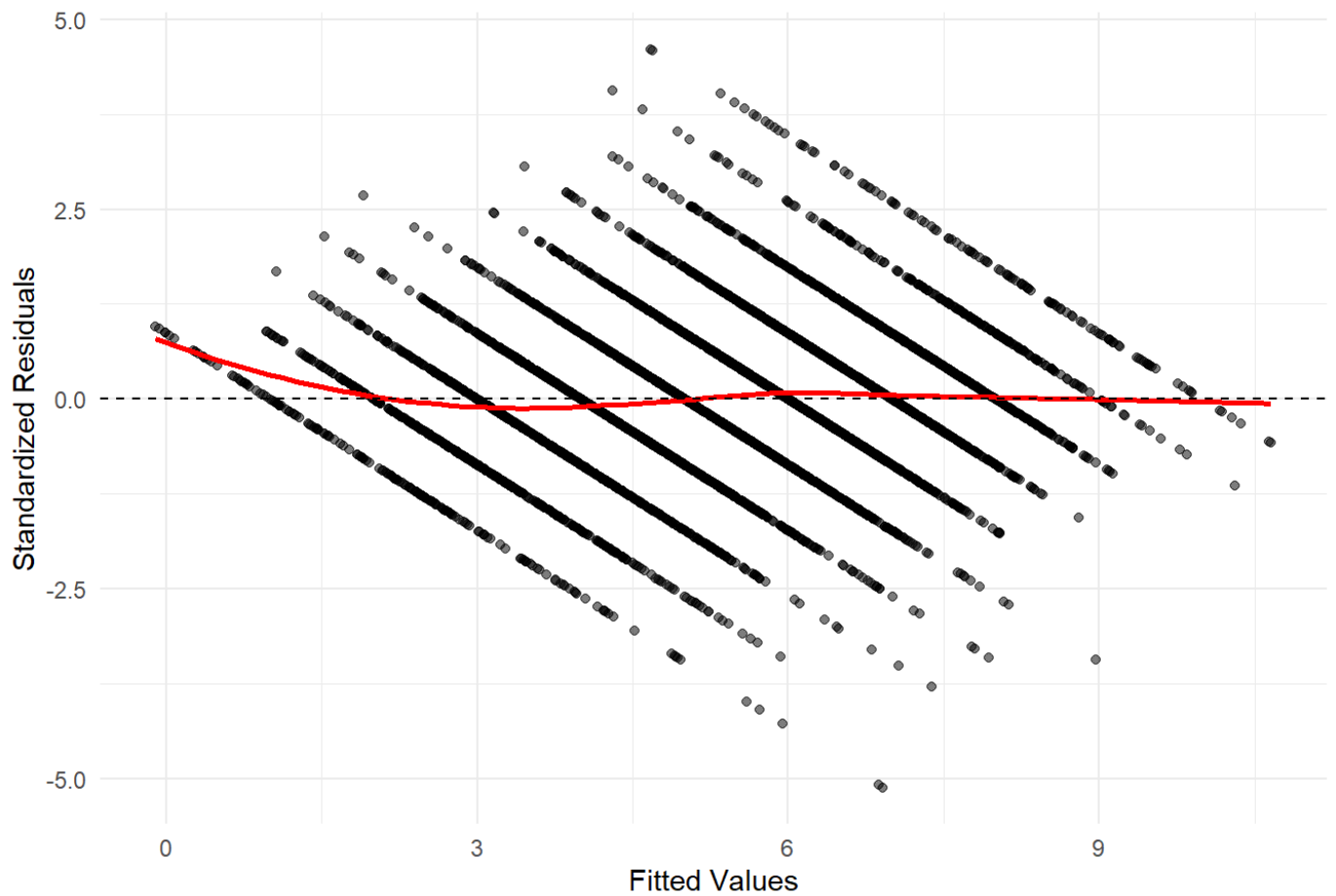


Figure B5.3. Standardized residuals versus fitted values for attractiveness ratings (image stimuli).

Red loess curve and horizontal reference line assess homoscedasticity assumption.

No systematic patterns indicate appropriate model fit.

Random Effects Q-Q: Attractiveness, Image - Model:block

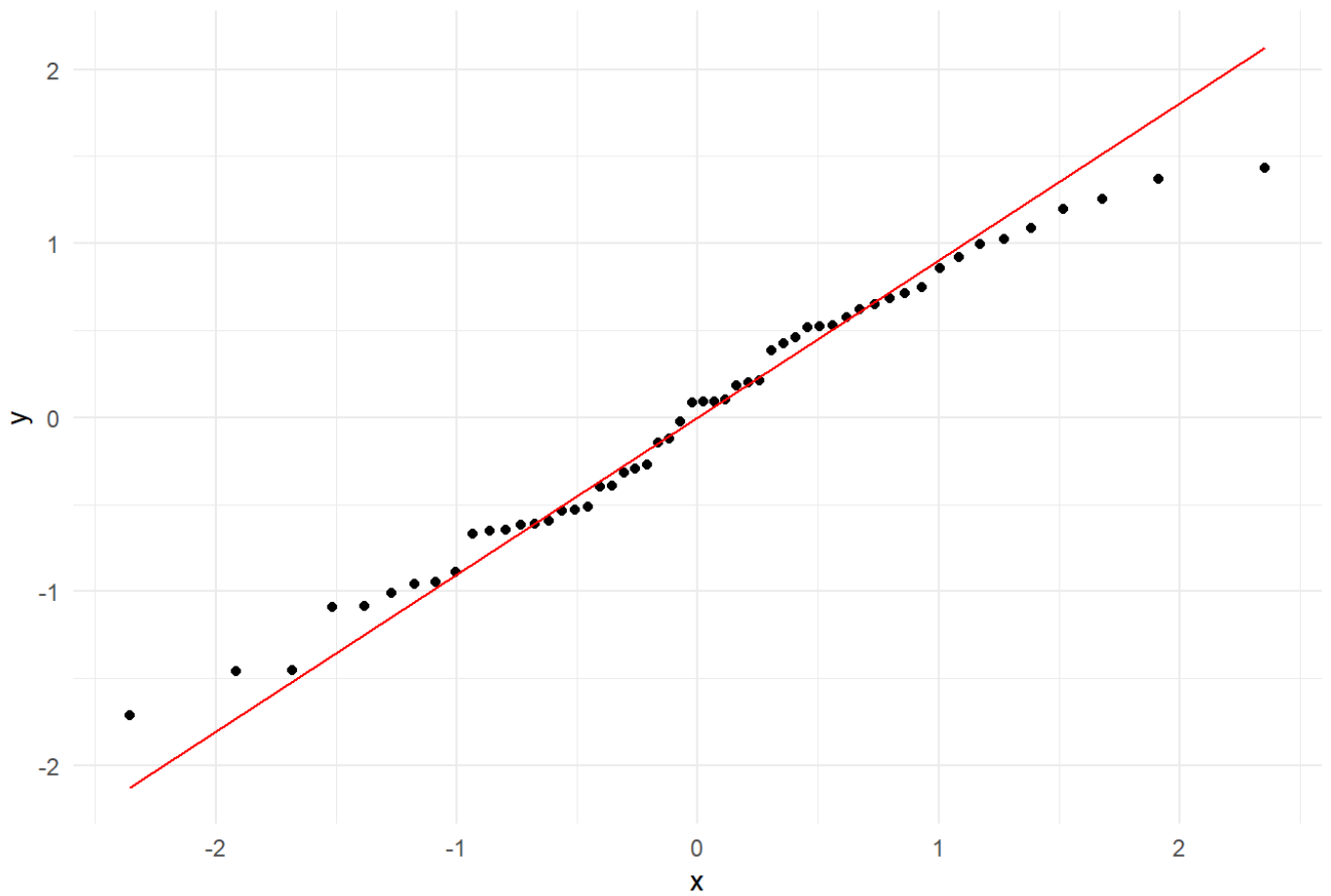


Figure B5.4. Q-Q plot of random intercepts for Model:block grouping factor (attractiveness, image stimuli).

Assesses normality assumption for nested random effects structure.

Random Effects Q-Q: Attractiveness, Image - code

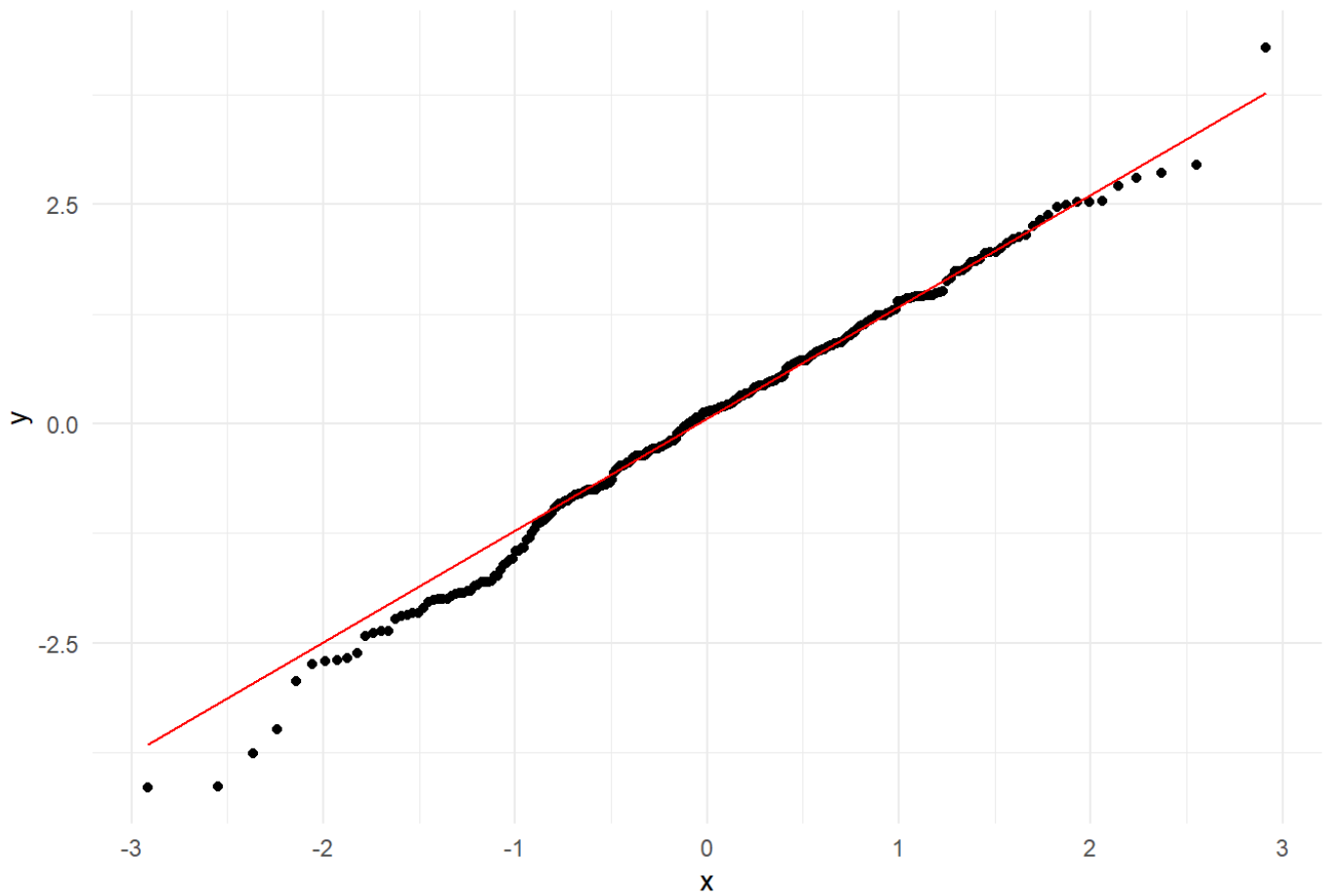
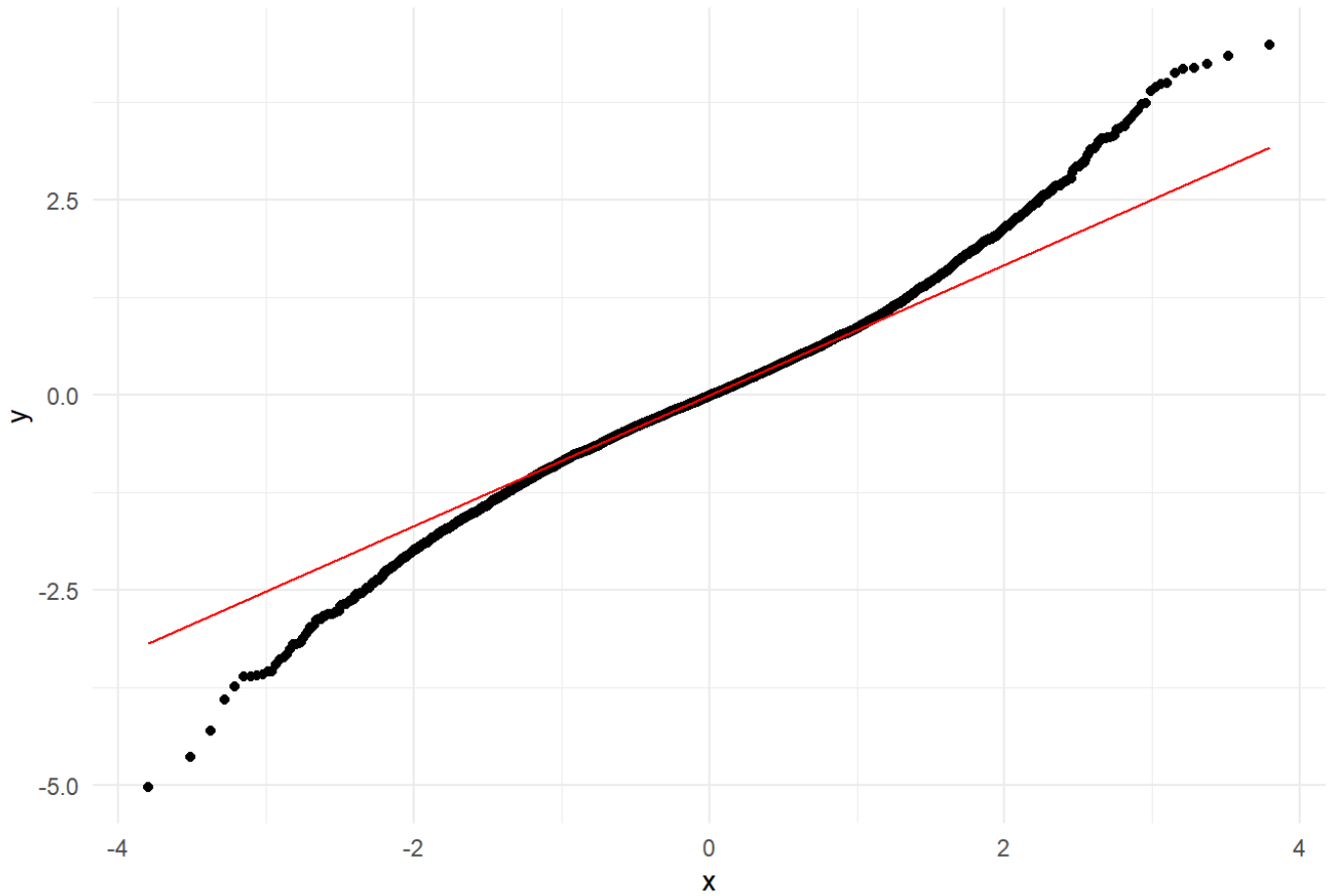


Figure B5.5. Q-Q plot of random intercepts for rater (code) grouping factor (attractiveness, image stimuli).

Assesses normality assumption for individual rater effects.

Q-Q Plot: Wealth, Image - Standardized Residuals



```
## Figure B5.6. Q-Q plot of standardized residuals for wealth perception ratings (image stimuli).  
## Points closely following the reference line (red),  
## indicate approximate normality of residuals.
```

Histogram: Wealth, Image - Standardized Residuals

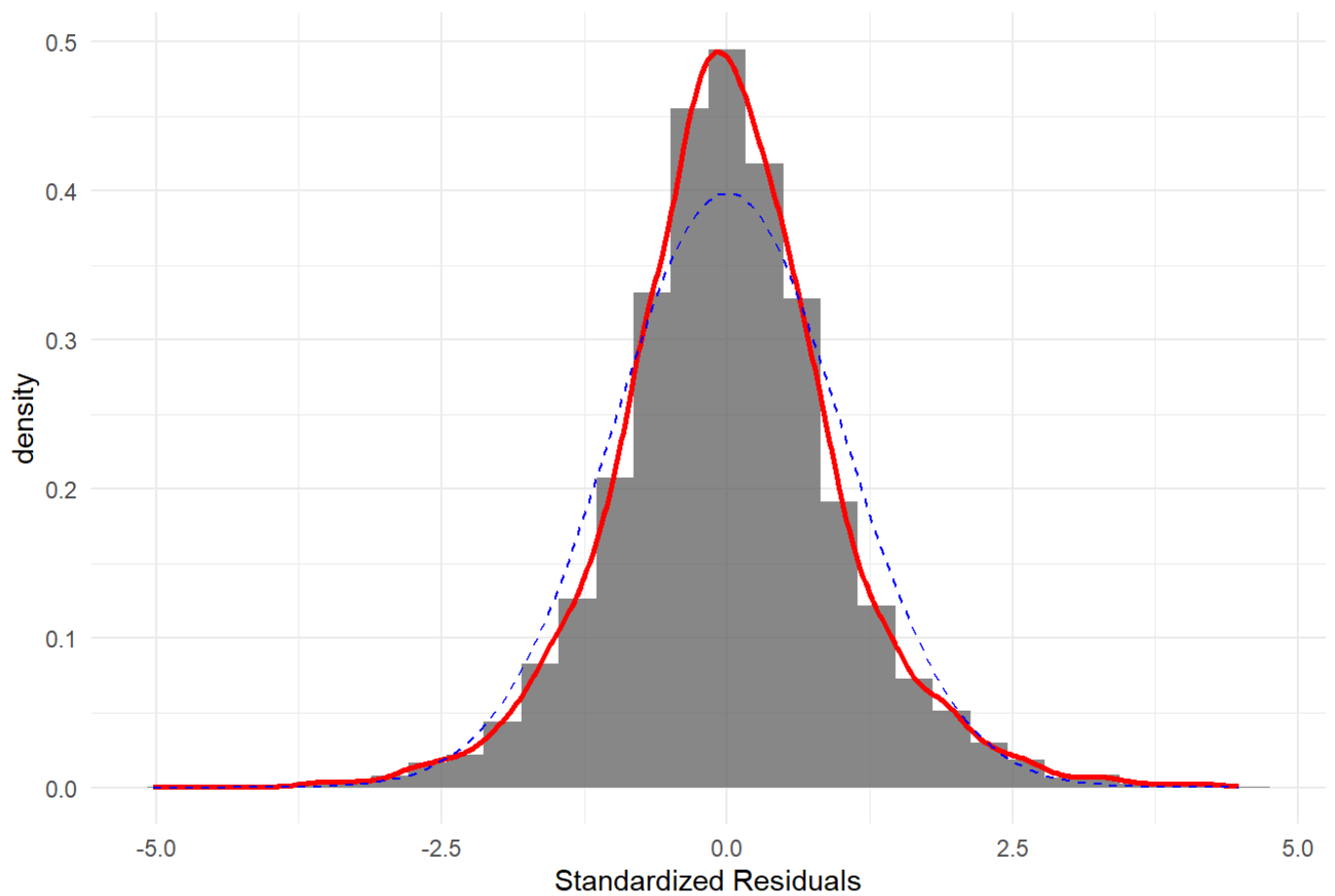


Figure B5.7. Histogram of standardized residuals for wealth perception ratings (image stimuli).

Red line shows kernel density estimate;

blue dashed line shows theoretical normal distribution ($\mu=0$, $\sigma=1$).

Residuals vs Fitted: Wealth, Image

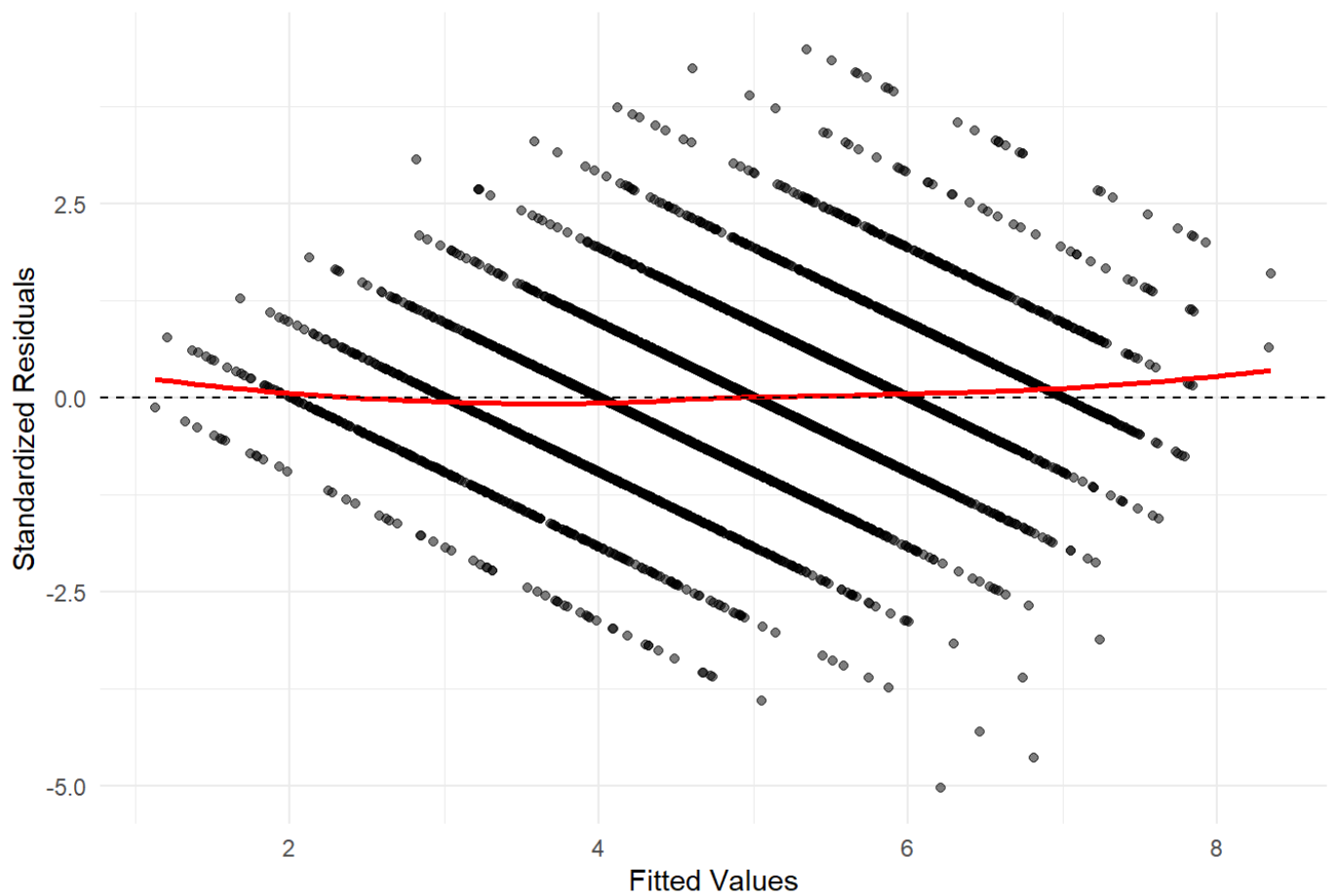


Figure B5.8. Standardized residuals versus fitted values for wealth perception ratings (image stimuli).

Red loess curve and horizontal reference line assess homoscedasticity assumption.

Random Effects Q-Q: Wealth, Image - Model:block

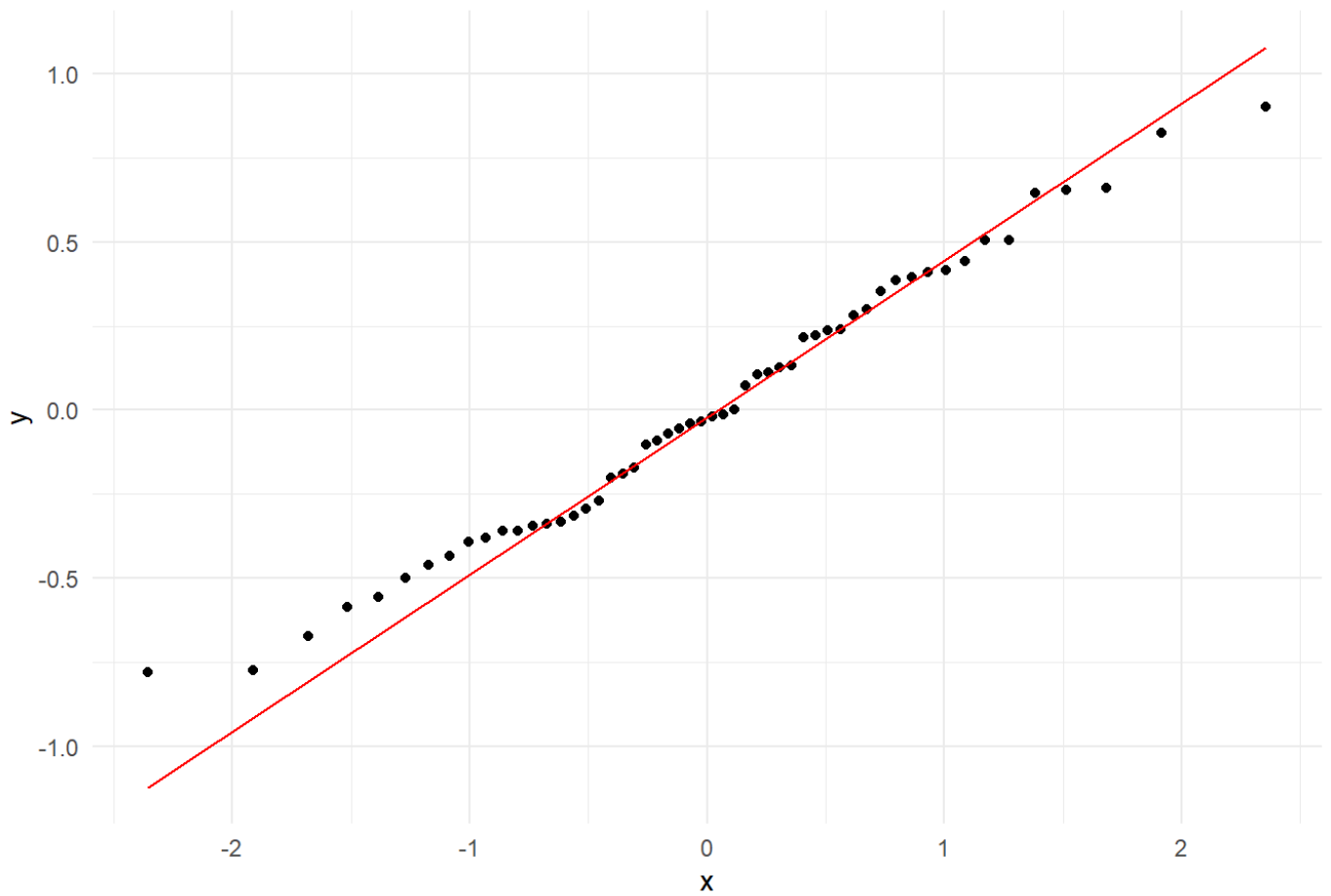
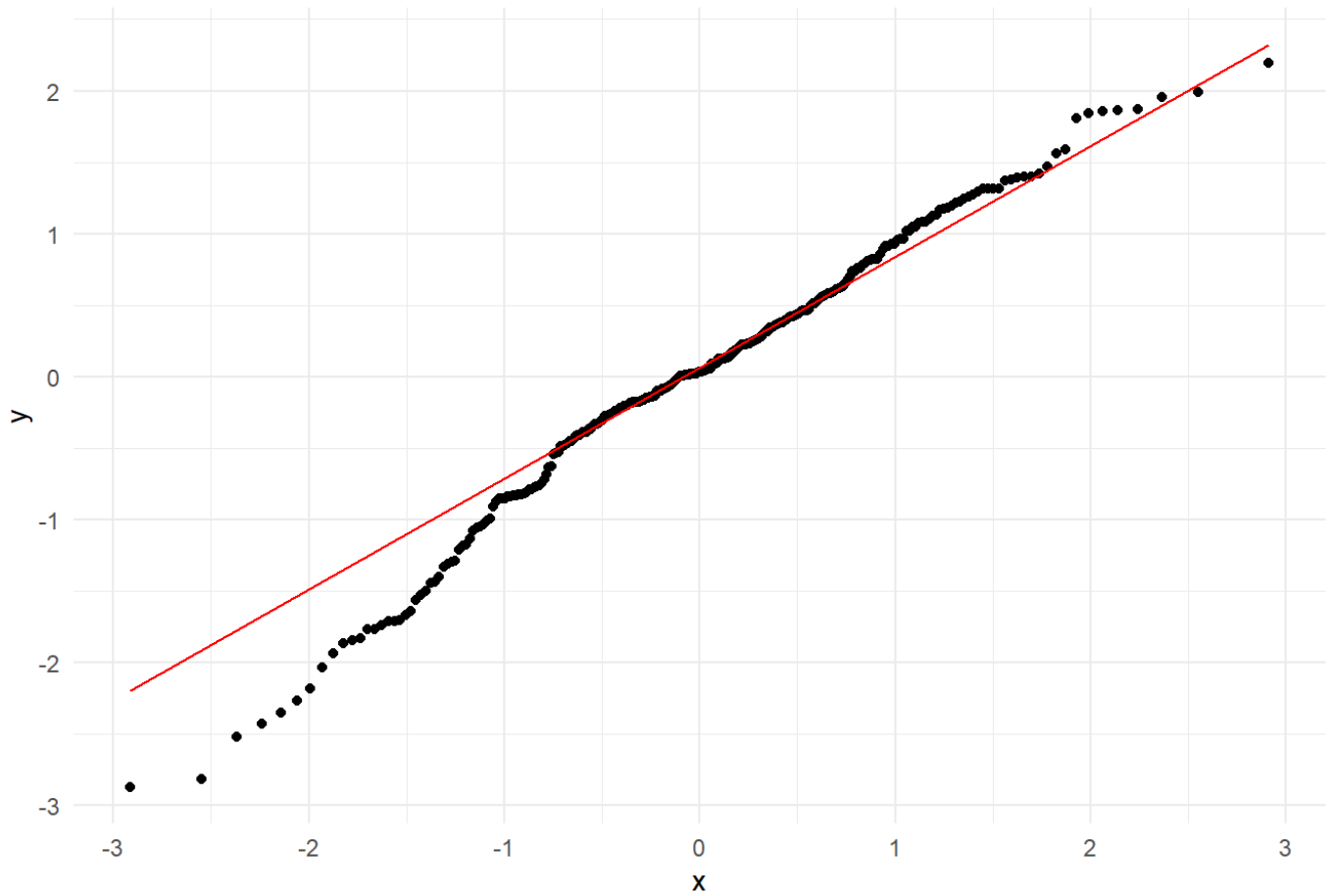


Figure B5.9. Q-Q plot of random intercepts for Model:block grouping factor (wealth, image stimuli).

Assesses normality assumption for nested random effects structure.

Random Effects Q-Q: Wealth, Image - code



```
## Figure B5.10. Q-Q plot of random intercepts for rater (code) grouping factor (wealth, image stimuli).
```

```
##     Assesses normality assumption for individual rater effects.
```

25. Diagnostic for Boundary fits and random effects variance

```
## === RANDOM EFFECTS EXAMINATION: Attractiveness ===
## WARNING: Model has singular/boundary fit - some variance components are zero
## Variance Components:
##   Groups      Name      Std.Dev.
##   code        (Intercept) 1.41033
##   Model:block (Intercept) 0.82282
##   block        (Intercept) 0.00000
##   Residual                1.15600
##
## Random Effects for code :
##   Number of levels: 279
##   Range: -4.1435 4.2883
##   Mean: 0
##   SD: 1.3754
##   Variance: 1.891649
##   Quick normality assessment:
##     Skewness: -0.24
##     Kurtosis: 3.116
##
## Random Effects for Model:block :
##   Number of levels: 54
##   Range: -1.7143 1.4323
##   Mean: 0
##   SD: 0.7882
##   Variance: 0.621337
##   Quick normality assessment:
##     Skewness: -0.114
##     Kurtosis: 2.168
##
## Random Effects for block :
##   Number of levels: 11
##   Range: 0 0
##   Mean: 0
##   SD: 0
##   Variance: 0
##   ** Near-zero variance detected (boundary fit)
```

```

## === RANDOM EFFECTS EXAMINATION: Wealth ===
## WARNING: Model has singular/boundary fit - some variance components are zero
## Variance Components:
##   Groups      Name      Std.Dev.
##   code      (Intercept) 9.6967e-01
##   Model:block (Intercept) 4.3990e-01
##   block      (Intercept) 3.9763e-05
##   Residual              1.0379e+00
##
## Random Effects for code :
##   Number of levels: 279
##   Range: -2.8667 2.1965
##   Mean: 0
##   SD: 0.9389
##   Variance: 0.881581
##   Quick normality assessment:
##     Skewness: -0.427
##     Kurtosis: 3.241
##
## Random Effects for Model:block :
##   Number of levels: 54
##   Range: -0.7785 0.9024
##   Mean: 0
##   SD: 0.4107
##   Variance: 0.168686
##   Quick normality assessment:
##     Skewness: 0.149
##     Kurtosis: 2.305
##
## Random Effects for block :
##   Number of levels: 11
##   Range: 0 0
##   Mean: 0
##   SD: 0
##   Variance: 0
##   ** Near-zero variance detected (boundary fit)

```

Residuals by Block/Model: Attractiveness

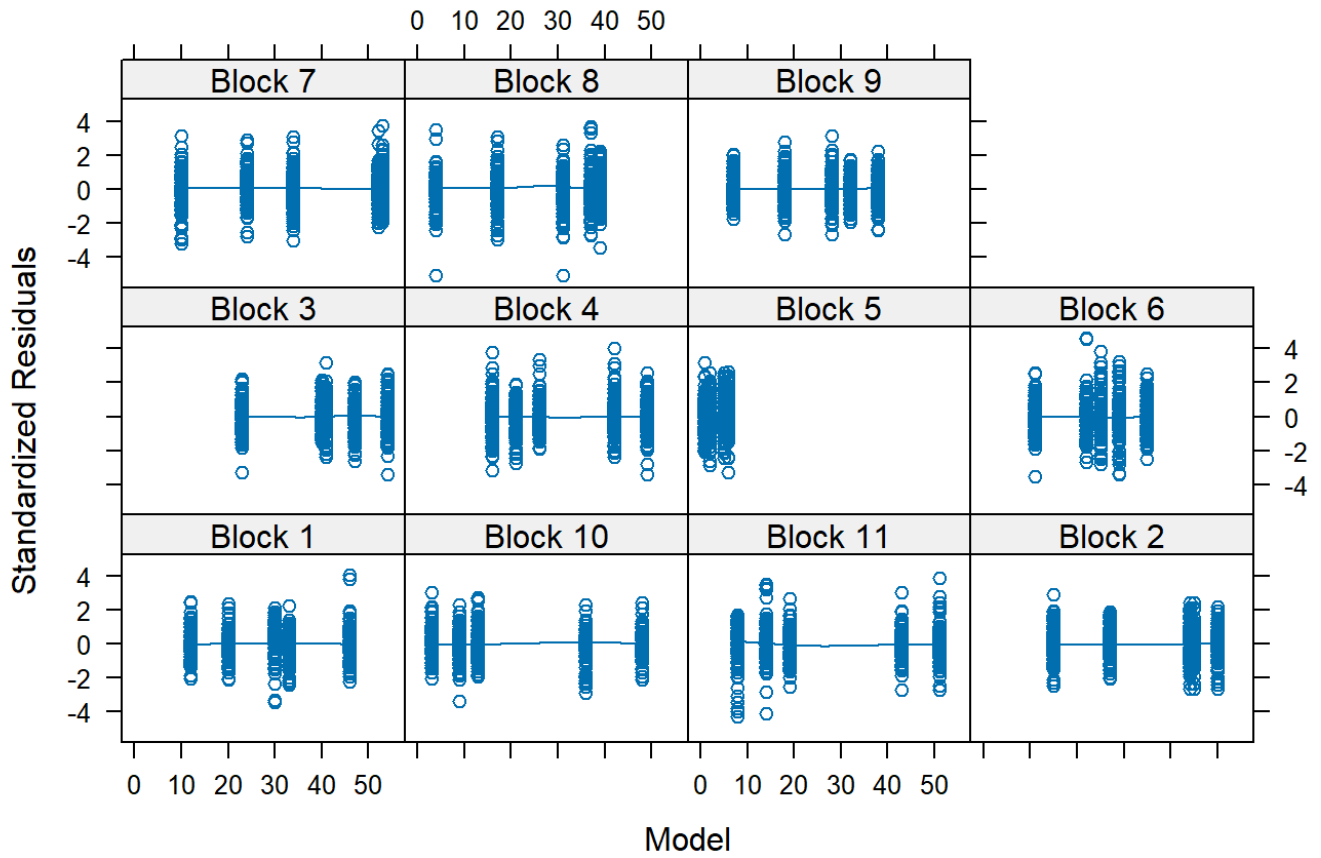
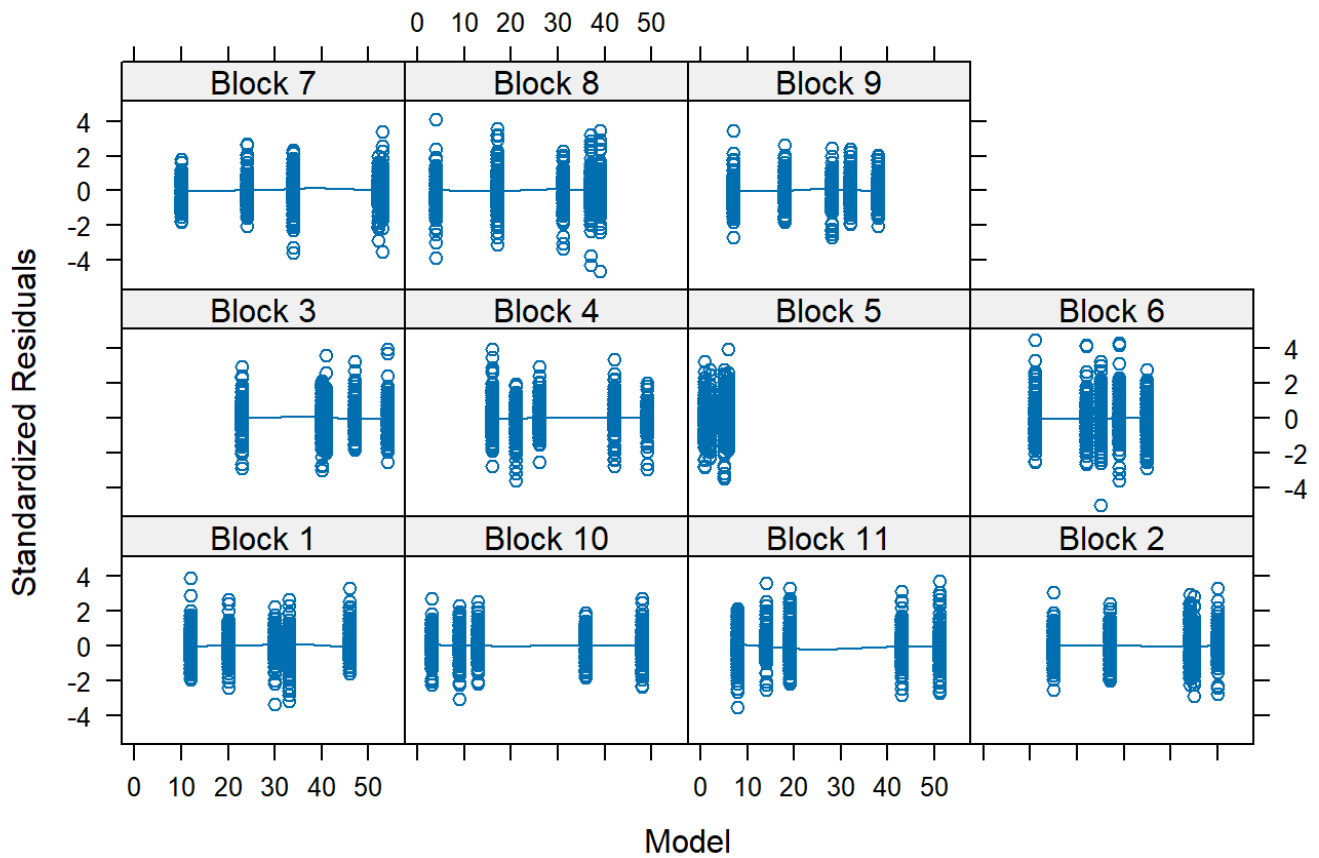


Figure B5.11. Standardized residuals by nested structure (Block/Model) for attractiveness ratings.

Panel plot shows residual patterns across experimental blocks

with loess smoothing curves to detect systematic variation.

Residuals by Block/Model: Wealth



```
## Figure B5.12. Standardized residuals by nested structure (Block/Model) for wealth percepti
on ratings.
##     Panel plot shows residual patterns across experimental blocks
##     with loess smoothing curves to detect systematic variation.
```

PART VI: Supplement section B6: Power

Analysis

Stimulus-Level Power Sensitivity Analysis

[illegible]

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
```

```
## #####(5/5) Simu
lating: |=====|
|#####(5/5) #####
```

```
## Power for predictor 'Condition', (95% confidence interval),
## by number of levels in Model:
##      20:  0.00% ( 0.00,  0.37) - 1987 rows
##      30:  0.00% ( 0.00,  0.37) - 3025 rows
##      40:  0.00% ( 0.00,  0.37) - 4035 rows
##      50:  0.00% ( 0.00,  0.37) - 5029 rows
##      NA:  0.00% ( 0.00,  0.37) - 5433 rows
##
## Time elapsed: 0 h 16 m 38 s
```

Rater level sample power analysis

2x2 Factorial Design: Trait_Exaggeration x Cost

```
##
## =====
## MODEL: ATTRACTIVENESS - IMAGE STIMULI
## =====
```

```
##
## =====
## MODEL: WEALTH - IMAGE STIMULI
## =====
```

```
##
## =====
```

STEP 2: ACHIEVED POWER ANALYSIS

```
## (Power achieved with observed effect sizes)
```

```
## =====
```

```
##
## Calculating achieved power for: ATTRACTIVENESS - IMAGE
## Number of simulations: 1000
## This may take a few minutes...
##
## Analyzing: Main effect of Trait_Exaggeration...
## Simulating: |
```

```
## [REDACTED]Simulation
g: | = |
```

```
## #####Simulatin
g: |== |
```

```
## [REDACTED]Simulating  
g: |==|
```

```
## #####Simulation
g: |===|
```

```
## #####Simulatin
g: |====|
```

```
## #####Simulating: |=====
|#####
```

```
##
## Trait_Exaggeration Power Results:
## Power for predictor 'Trait_Exaggeration1', (95% confidence interval):
##     100.0% (99.63, 100.0)
##
## Test: z-test
##     Effect size for Trait_Exaggeration1 is -0.16
##
## Based on 1000 simulations, (56 warnings, 0 errors)
## alpha = 0.05, nrow = 5450
##
## Time elapsed: 0 h 5 m 31 s
##
## nb: result might be an observed power calculation
##
## Analyzing: Main effect of Cost...
## Simulating: |
```

```
## #####Simulating: |=
```

```
## #####Simulating: |==
```

```
## #####Simulating: |===
```

```
## #####Simulating: |====
```

```
## #####Simulating: |=====
```

```
## #####Simulating: |=====
```

```
## #####Simulating: |=====
```

```
## #####Simulating: |=====
```

```
##  
## Cost Power Results:  
## Power for predictor 'Cost1', (95% confidence interval):  
##      21.80% (19.28, 24.49)  
##  
## Test: z-test  
##      Effect size for Cost1 is -0.018  
##  
## Based on 1000 simulations, (54 warnings, 0 errors)  
## alpha = 0.05, nrow = 5450  
##  
## Time elapsed: 0 h 5 m 27 s  
##  
## nb: result might be an observed power calculation
```

```
##
## Calculating achieved power for: WEALTH - IMAGE
## Number of simulations: 1000
## This may take a few minutes...
##
## Analyzing: Main effect of Trait_Exaggeration...
## Simulating: |
```

```
## #####Simulatin
g: | = |
```

```
## #####Simulatin
g: |== |
```

```
## #####Simulatin
g: |==|
```

```
## #####Simulatin
g: |===|
```

[illegible]

```
## #####Simulatin
g: |=====|
```

```
## #####Simulatin
g: |=====|
```

[illegible]


```
##
## Cost Power Results:
## Power for predictor 'Cost1', (95% confidence interval):
##      94.60% (93.01, 95.92)
##
## Test: z-test
##      Effect size for Cost1 is -0.050
##
## Based on 1000 simulations, (31 warnings, 0 errors)
## alpha = 0.05, nrow = 5433
##
## Time elapsed: 0 h 5 m 30 s
##
## nb: result might be an observed power calculation
```

```
##
## =====
```

```
## STEP 3: MINIMUM DETECTABLE EFFECT SIZE ANALYSIS
```

```
## (Smallest effect detectable with 80% power)
```

```
## =====
```

```
## -----
```

```
## ATTRACTIVENESS MODEL - MDES Analysis
```

```
## -----
```

```
##
## Finding MDES for: Trait_Exaggeration1 in ATTRACTIVENESS - Trait_Exaggeration
## Target power: 0.8
## This will test multiple effect sizes...
##
## Observed effect size: -0.1625
## Testing effect size:-0.0325(20% of observed)...
```

```
## Power:0.52 [0.448, 0.591]
## Testing effect size:-0.065(40% of observed)...
```

```
## Power:0.985 [0.957, 0.997]
## Testing effect size:-0.0975(60% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.13(80% of observed)...
```



```
## Power:1 [0.982, 1]
## Testing effect size:-0.1625(100% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.195(120% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.2275(140% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.26(160% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.2925(180% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.325(200% of observed)...
```

```
## Power:1 [0.982, 1]
##
## ** MINIMUM DETECTABLE EFFECT SIZE **
## Effect size for ~80% power: -0.065
## Achieved power at this effect size: 0.985
## This is 40% of the observed effect
```

```
##
## Finding MDES for: Cost1 in ATTRACTIVENESS - Cost
## Target power: 0.8
## This will test multiple effect sizes...
##
## Observed effect size: -0.0182
## Testing effect size:-0.0036(20% of observed)...
```

```
## Power:0.045 [0.021, 0.084]
## Testing effect size:-0.0073(40% of observed)...
```

```
## Power:0.085 [0.05, 0.133]
## Testing effect size:-0.0109(60% of observed)...
```

```
## Power:0.105 [0.066, 0.156]
## Testing effect size:-0.0145(80% of observed)...
```

```
## Power:0.17 [0.121, 0.229]
## Testing effect size:-0.0182(100% of observed)...
```

```
## Power:0.215 [0.16, 0.278]
## Testing effect size:-0.0218(120% of observed)...
```

```
## Power:0.25 [0.192, 0.316]
## Testing effect size:-0.0254(140% of observed)...
```

```
## Power:0.325 [0.261, 0.395]
## Testing effect size:-0.0291(160% of observed)...
```

```
## Power:0.405 [0.336, 0.477]
## Testing effect size:-0.0327(180% of observed)...
```

```
## Power:0.535 [0.463, 0.606]
## Testing effect size:-0.0363(200% of observed)...
```

```
## Power:0.61 [0.539, 0.678]
##
## ** MINIMUM DETECTABLE EFFECT SIZE **
## Effect size for ~80% power: -0.0363
## Achieved power at this effect size: 0.61
## This is 200% of the observed effect
```

```
## -----
```

```
## WEALTH MODEL - MDES Analysis
```

```
## -----
```

```
##
## Finding MDES for: Trait_Exaggeration1 in WEALTH - Trait_Exaggeration
## Target power: 0.8
## This will test multiple effect sizes...
##
## Observed effect size: -0.252
## Testing effect size:-0.0504(20% of observed)...
```

```
## Power:0.965 [0.929, 0.986]
## Testing effect size:-0.1008(40% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.1512(60% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.2016(80% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.252(100% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.3024(120% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.3528(140% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.4032(160% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.4536(180% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.504(200% of observed)...
```

```
## Power:1 [0.982, 1]
##
## ** MINIMUM DETECTABLE EFFECT SIZE **
## Effect size for ~80% power: -0.0504
## Achieved power at this effect size: 0.965
## This is 20% of the observed effect
```

```
##
## Finding MDES for: Cost1 in WEALTH - Cost
## Target power: 0.8
## This will test multiple effect sizes...
##
## Observed effect size: -0.05
## Testing effect size:-0.01(20% of observed)...
```

```
## Power:0.125 [0.083, 0.179]
## Testing effect size:-0.02(40% of observed)...
```

```
## Power:0.36 [0.294, 0.431]
## Testing effect size:-0.03(60% of observed)...
```

```
## Power:0.61 [0.539, 0.678]
## Testing effect size:-0.04(80% of observed)...
```

```
## Power:0.86 [0.804, 0.905]
## Testing effect size:-0.05(100% of observed)...
```

```
## Power:0.97 [0.936, 0.989]
## Testing effect size:-0.06(120% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.0701(140% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.0801(160% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.0901(180% of observed)...
```

```
## Power:1 [0.982, 1]
## Testing effect size:-0.1001(200% of observed)...
```

```
## Power:1 [0.982, 1]
##
## ** MINIMUM DETECTABLE EFFECT SIZE **
## Effect size for ~80% power: -0.04
## Achieved power at this effect size: 0.86
## This is 80% of the observed effect
```

```
##
## =====
```

```
## STEP 4: CREATING POWER CURVE VISUALIZATIONS
```

```
## =====
```

```
## Saved plot: power_curve_attractiveness_trait.png
```

```
## Saved plot: power_curve_attractiveness_cost.png
```

```
## Saved plot: power_curve_wealth_trait.png
```

```
## Saved plot: power_curve_wealth_cost.png
```

```
##
## =====
```

```
## FINAL SUMMARY REPORT
```

```
## =====
```

```
##
## ATTRACTIVENESS MODEL
## -----
##                               Effect Observed_Effect Achieved_Power
## Trait_Exaggeration1 Trait_Exaggeration      -0.1625      1.000
## Cost1                               Cost      -0.0182      0.218
##                               MDES_80pct_Power MDES_as_pct_of_observed
## Trait_Exaggeration1      -0.0650                               40
## Cost1                    -0.0363                               200
```

```
##
## WEALTH MODEL
## -----
##                               Effect Observed_Effect Achieved_Power
## Trait_Exaggeration1 Trait_Exaggeration      -0.252      1.000
## Cost1                               Cost      -0.050      0.946
##                               MDES_80pct_Power MDES_as_pct_of_observed
## Trait_Exaggeration1      -0.0504                               20
## Cost1                    -0.0400                               80
```

PART VII: Supplement B7: Expected versus Observed Ratings of Isolated Item Photographs

At the end of each rating session, participants additionally viewed standardised close-up photographs of the four experimental footwear items (Economy Flat, Economy Heel, Luxury Flat, Luxury Heel), presented independently of any model, and rated on the same Likert scales responding to estimate the appearance of attractiveness and wealth of an average hypothetical user. These isolated item ratings provide an **Expected** score, which can be contrasted with the **Observed** score obtained when the corresponding item was actually worn by a stimulus model. This section reports the data preparation, comparison, and inferential testing of these Expected versus Observed difference scores.

The Expected and Observed difference scores are defined as follows:

Expected [per item] = Isolated item-photograph rating – Control condition mean (image stimuli)

Observed [per condition] = Worn-item condition mean rating – Control condition mean

Both scores are referenced against the same control (no footwear) baseline, so that a divergence between Expected and Observed indicates that the isolated appraisal of an item over- or under-predicts its effect once embodied by a person.

26. Data Preparation for the Expected versus Observed Comparison

27. Paired-Samples Comparisons: Expected versus Observed Differences (Overall Sample)

Paired-samples *t*-tests were conducted for each condition, separately for the Attractiveness and Wealth dimensions and for image and video stimuli, testing whether the Expected difference score (isolated item photograph) differed statistically significantly from the Observed difference score (worn item).

Attractiveness (Image Stimuli): Expected vs. Observed Differences

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-11.200	262	0.0000	-1.455	***
EH.t	Economy Heel	2.065	262	0.0399	0.315	*
LF.t	Luxury Flat	-8.903	262	0.0000	-1.236	***
LH.t	Luxury Heel	2.786	262	0.0057	0.470	**

Attractiveness (Video Stimuli): Expected vs. Observed Differences

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-8.917	262	0	-1.202	***
EH.t	Economy Heel	6.260	262	0	0.983	***
LF.t	Luxury Flat	-6.194	262	0	-0.832	***
LH.t	Luxury Heel	8.471	262	0	1.455	***

Wealth (Image Stimuli): Expected vs. Observed Differences

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-7.469	262	0.0000	-0.891	***
EH.t	Economy Heel	4.342	262	0.0000	0.543	***
LF.t	Luxury Flat	-0.965	262	0.3352	-0.130	ns
LH.t	Luxury Heel	10.046	262	0.0000	1.401	***

Wealth (Video Stimuli): Expected vs. Observed Differences

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-7.558	262	0.0000	-0.910	***
EH.t	Economy Heel	4.472	262	0.0000	0.598	***
LF.t	Luxury Flat	-0.735	262	0.4629	-0.099	ns
LH.t	Luxury Heel	11.224	262	0.0000	1.711	***

28. Effect Sizes (Cohen's d) for Expected versus Observed Comparisons

Effect Sizes: Attractiveness (Image Stimuli)

Condition	Cohens_d
Economy Flat	-0.691
Economy Heel	0.127
Luxury Flat	-0.549

Condition	Cohens_d
Luxury Heel	0.172

Effect Sizes: Attractiveness (Video Stimuli)

Condition	Cohens_d
Economy Flat	-0.550
Economy Heel	0.386
Luxury Flat	-0.382
Luxury Heel	0.522

Effect Sizes: Wealth (Image Stimuli)

Condition	Cohens_d
Economy Flat	-0.461
Economy Heel	0.268
Luxury Flat	-0.060
Luxury Heel	0.619

Effect Sizes: Wealth (Video Stimuli)

Condition	Cohens_d
Economy Flat	-0.466
Economy Heel	0.276
Luxury Flat	-0.045
Luxury Heel	0.692

29. Paired-Samples Comparisons Stratified by Sex

Female Participants - Attractiveness (Image Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-9.057	163	0e+00	-1.543	***
EH.t	Economy Heel	4.071	163	1e-04	0.769	***
LF.t	Luxury Flat	-6.183	163	0e+00	-1.102	***
LH.t	Luxury Heel	5.212	163	0e+00	1.061	***

Female Participants - Attractiveness (Video Stimuli)

Condition	t_value	df	p_value	mean_diff	significance
-----------	---------	----	---------	-----------	--------------

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-7.810	163	0	-1.334	***
EH.t	Economy Heel	7.575	163	0	1.474	***
LF.t	Luxury Flat	-4.204	163	0	-0.724	***
LH.t	Luxury Heel	9.891	163	0	2.108	***

Female Participants - Wealth (Image Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-7.419	163	0.0000	-1.134	***
EH.t	Economy Heel	2.981	163	0.0033	0.482	**
LF.t	Luxury Flat	-0.843	163	0.4004	-0.145	ns
LH.t	Luxury Heel	8.368	163	0.0000	1.493	***

Female Participants - Wealth (Video Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-7.570	163	0.0000	-1.156	***
EH.t	Economy Heel	3.658	163	0.0003	0.652	***
LF.t	Luxury Flat	-0.936	163	0.3509	-0.158	ns
LH.t	Luxury Heel	9.886	163	0.0000	1.954	***

Male Participants - Attractiveness (Image Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-6.583	98	0.0000	-1.308	***
EH.t	Economy Heel	-1.812	98	0.0731	-0.436	ns
LF.t	Luxury Flat	-6.617	98	0.0000	-1.457	***
LH.t	Luxury Heel	-1.887	98	0.0621	-0.508	ns

Male Participants - Attractiveness (Video Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-4.493	98	0.0000	-0.982	***
EH.t	Economy Heel	0.688	98	0.4932	0.168	ns
LF.t	Luxury Flat	-4.720	98	0.0000	-1.011	***
LH.t	Luxury Heel	1.461	98	0.1471	0.373	ns

Male Participants - Wealth (Image Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-2.646	98	0.0095	-0.488	**
EH.t	Economy Heel	3.267	98	0.0015	0.644	**
LF.t	Luxury Flat	-0.483	98	0.6305	-0.104	ns
LH.t	Luxury Heel	5.581	98	0.0000	1.248	***

Male Participants - Wealth (Video Stimuli)

	Condition	t_value	df	p_value	mean_diff	significance
EF.t	Economy Flat	-2.649	98	0.0094	-0.503	**
EH.t	Economy Heel	2.565	98	0.0118	0.509	*
LF.t	Luxury Flat	-0.009	98	0.9929	-0.002	ns
LH.t	Luxury Heel	5.592	98	0.0000	1.307	***

30. Mixed-Effects Models: Sex × Measure (Expected vs. Observed) Interactions

A linear mixed-effects model was fitted for each condition, stimulus type, and dimension, with Measure (Expected vs. Observed) and sex entered as fixed effects together with their interaction, and a random intercept for participant, to test whether the divergence between Expected and Observed ratings differed by sex.

Sex x Measure (Expected vs. Observed) Interaction Tests Across Conditions

Model	sex_p	measure_p	interaction_p
EF_Img_Att	0.8477	0.0000	0.3649
EF_Img_Wlt	0.0398	0.0000	0.0063
EF_Vid_Att	0.9211	0.0000	0.2146
EF_Vid_Wlt	0.0241	0.0000	0.0116
EH_Img_Att	0.0000	0.3161	0.0001
EH_Img_Wlt	0.2600	0.0000	0.6333
EH_Vid_Att	0.0000	0.0000	0.0000
EH_Vid_Wlt	0.7748	0.0000	0.4964
LF_Img_Att	0.1413	0.0000	0.2179
LF_Img_Wlt	0.9256	0.3619	0.8826
LF_Vid_Att	0.1219	0.0000	0.2925
LF_Vid_Wlt	0.8750	0.5287	0.6211
LH_Img_Att	0.0000	0.1094	0.0000

Model	sex_p	measure_p	interaction_p
LH_Img_Wlt	0.0642	0.0000	0.3489
LH_Vid_Att	0.0000	0.0000	0.0000
LH_Vid_Wlt	0.5304	0.0000	0.0227

31. Descriptive Statistics for Expected and Observed Difference Scores

Descriptives: Attractiveness (Image Stimuli)

Variable	Mean	SD	Median	Min	Max
Exp_EF_Att	-1.48	2.27	-1.6	-6.6	5.00
Obs_EF_Img_Att	-0.03	0.64	0.0	-3.6	2.40
Exp_EH_Att	0.64	2.73	0.8	-6.2	8.75
Obs_EH_Img_Att	0.35	0.80	0.2	-2.0	4.00
Exp_LF_Att	-1.17	2.33	-1.2	-7.4	4.20
Obs_LF_Img_Att	0.06	0.59	0.0	-2.8	2.40
Exp_LH_Att	0.78	2.98	1.0	-6.6	8.00
Obs_LH_Img_Att	0.33	0.83	0.2	-2.4	4.20

Descriptives: Attractiveness (Video Stimuli)

Variable	Mean	SD	Median	Min	Max
Exp_EF_Att	-1.48	2.27	-1.6	-6.6	5.00
Obs_EF_Vid_Att	-0.26	0.80	-0.2	-3.4	3.60
Exp_EH_Att	0.64	2.73	0.8	-6.2	8.75
Obs_EH_Vid_Att	-0.31	1.15	-0.2	-4.0	4.75
Exp_LF_Att	-1.17	2.33	-1.2	-7.4	4.20
Obs_LF_Vid_Att	-0.32	0.79	-0.2	-2.8	3.40
Exp_LH_Att	0.78	2.98	1.0	-6.6	8.00
Obs_LH_Vid_Att	-0.64	1.23	-0.6	-5.0	5.50

Descriptives: Wealth (Image Stimuli)

Variable	Mean	SD	Median	Min	Max
Exp_EF_Wlt	-0.69	2.10	-0.80	-5.40	5.00
Obs_EF_Img_Wlt	0.20	0.71	0.05	-2.80	2.20
Exp_EH_Wlt	1.30	2.37	1.20	-4.75	7.00

Variable	Mean	SD	Median	Min	Max
Obs_EH_Img_Wlt	0.79	1.03	0.60	-2.00	5.80
Exp_LF_Wlt	0.25	2.32	0.40	-6.20	6.20
Obs_LF_Img_Wlt	0.38	0.70	0.25	-1.80	3.25
Exp_LH_Wlt	2.17	2.62	2.20	-4.75	8.00
Obs_LH_Img_Wlt	0.81	1.01	0.60	-3.00	5.00

Descriptives: Wealth (Video Stimuli)

Variable	Mean	SD	Median	Min	Max
Exp_EF_Wlt	-0.69	2.10	-0.8	-5.40	5.00
Obs_EF_Vid_Wlt	0.22	0.82	0.2	-2.00	4.25
Exp_EH_Wlt	1.30	2.37	1.2	-4.75	7.00
Obs_EH_Vid_Wlt	0.71	1.34	0.6	-2.50	7.20
Exp_LF_Wlt	0.25	2.32	0.4	-6.20	6.20
Obs_LF_Vid_Wlt	0.36	0.87	0.2	-2.20	4.00
Exp_LH_Wlt	2.17	2.62	2.2	-4.75	8.00
Obs_LH_Vid_Wlt	0.47	1.42	0.2	-3.25	8.00

32. Visualization of Expected versus Observed Differences

The following plots follow the visual conventions established in Section B4 (`theme_minimal` , bordered facet panels, top-aligned bold legend) to maintain a consistent style across the supplementary analyses.

Expected vs. Observed Differences from Control Condition

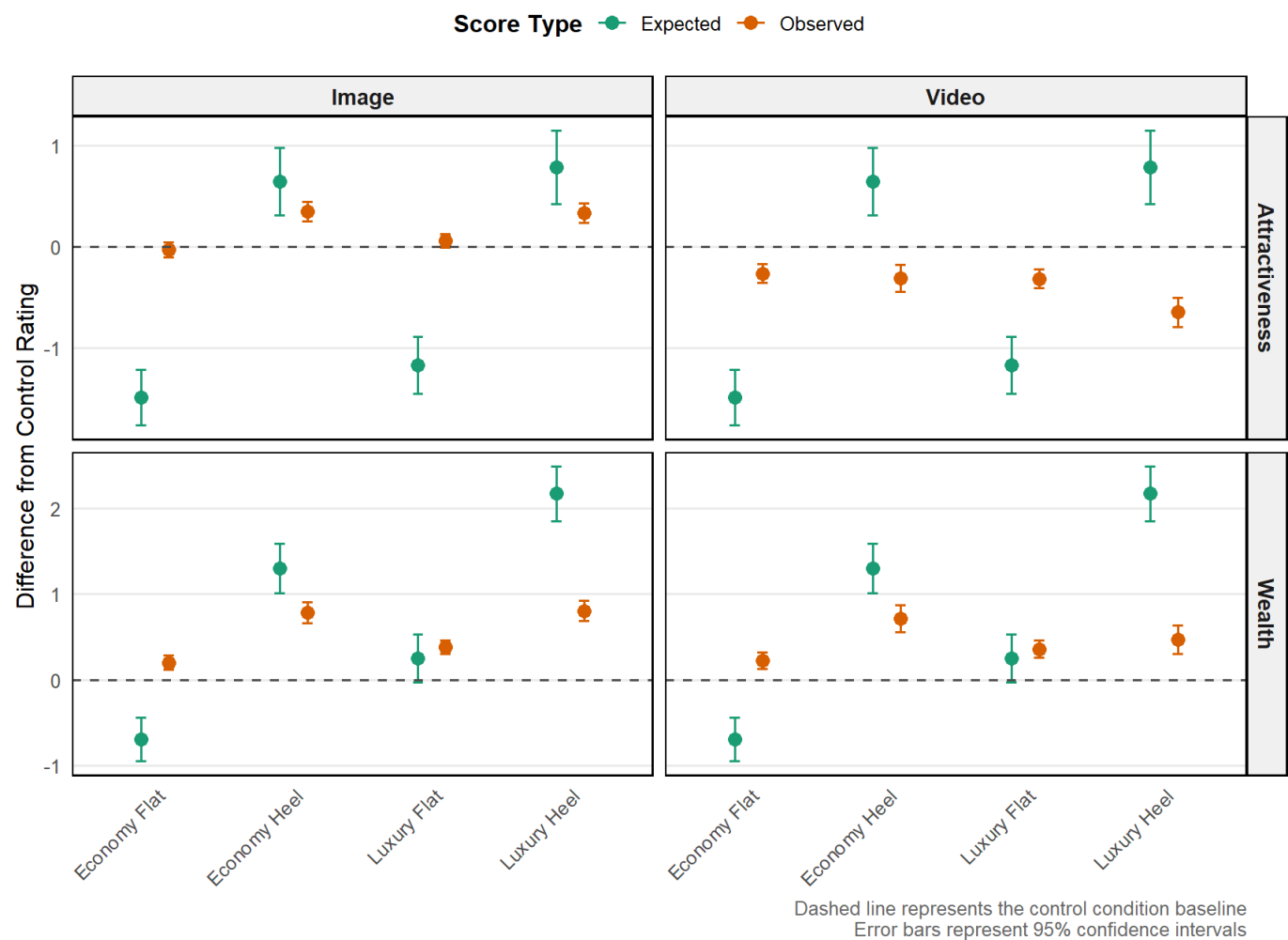


Figure B7.1. Expected (isolated item photograph) versus Observed (worn item) differences from the control condition, by footwear condition, rating dimension, and stimulus type. Error bars represent 95% confidence intervals.

Expected vs. Observed Differences from Control Condition, by Sex

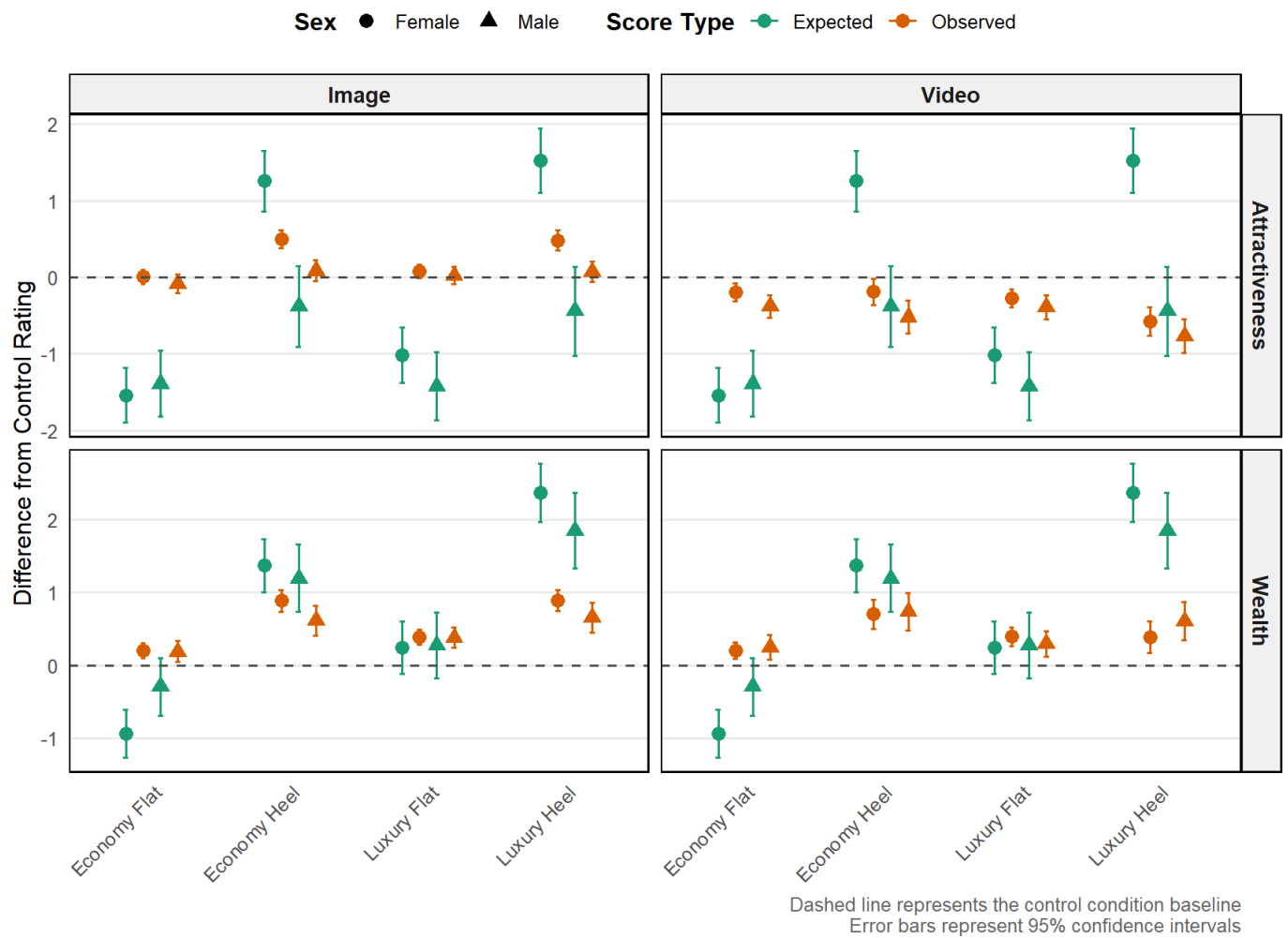


Figure B7.2. Expected versus Observed differences from the control condition, stratified by participant sex. Error bars represent 95% confidence intervals.